

Staggered Rollout for Innovation Adoption^{*}

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June 2026

Abstract

Adoption generates public information, which gives forward-looking agents an incentive to let others experiment first. A principal can counter this delay by committing to dated supply. Under free availability, strategic waiting produces convex adoption within payoff groups and, with heterogeneous payoffs, an aggregate S-shaped diffusion curve. Scarcity eliminates gradual adoption with one payoff type, and with two target-relevant types an initial and a terminal exhausted batch attain the unrestricted optimum. The boundary appears at three types: rejection-contingent rationing becomes a screening instrument, and four releases can strictly outperform every immediately exhausted plan with at most three release dates. For ordered continuum schedules in which almost every value receives one service date, the prior and target determine unique compulsory initial and terminal batch cores. Between them, every ordered rollout decomposes into smooth movement along an explicit indifference ridge and pooled excursions. For a positive nonincreasing density, every nondegenerate excursion delays completion, so the unique optimum in this class is an initial batch, smooth rollout, and terminal batch. If density instead rises sufficiently rapidly on an active region, pooling can be locally faster and, under a global replacement condition, an optimal rollout contains a wave. Thus distributional shape determines whether staggered access separates agents smoothly or advances through pooled waves.

Keywords: innovation adoption, social learning, scarcity, dynamic mechanism design, rationing, diffusion.

JEL codes: C73, D82, D83, O33.

1 Introduction

In December 2020, the first COVID-19 vaccines were authorized. Because supply was scarce, access expanded in waves: first health workers, then older people and patients with comorbidities, and eventually the general population. Medical risk and equity largely determined that ordering. Yet the waves also changed incentives. Each vaccinated cohort generated public information about safety and effectiveness, so someone expecting access later could wait for that evidence rather than experiment early.

^{*}I am especially grateful to Bobak Pakzad-Hurson for his guidance and continued support. I also thank Jack Fanning, Teddy Mekonnen, Kareen Rozen, Roberto Serrano, and audiences at Brown University, LACEA-LAMES, EWMES, SBE, USP, Universidad de Chile, and Universidad de los Andes for helpful comments. All remaining errors are mine.

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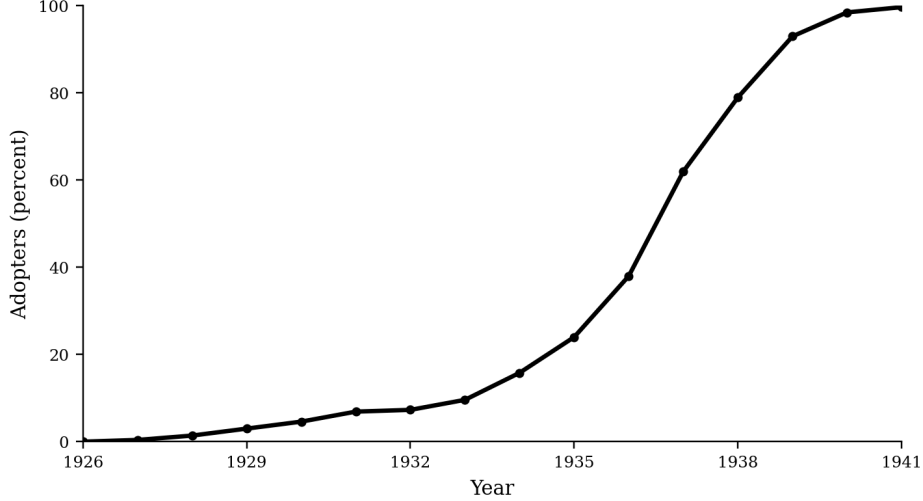


Figure 1: Hybrid-corn diffusion in two Iowa communities. Author’s reconstruction from the annual adoption shares reported in [Ryan and Gross \(1943, Figure 4\)](#). The model generates an S-shaped path from strategic delay and heterogeneity in the payoff from successful adoption rather than from myopic behavior.

The same tension arises whenever adoption reveals quality and can be postponed. Farmers may wait for neighbors to try a new seed, physicians for broader experience with an unfamiliar treatment, and firms for early users to reveal the profitability of a new technology. Digital platforms and artificial-intelligence providers likewise rely on invitations, waitlists, and capacity tiers when releasing new products. Across these settings, availability determines not only who can adopt, but also how valuable it is to wait.

The empirical diffusion literature has long emphasized social learning in agriculture, health, and technology adoption ([Ryan and Gross, 1943](#); [Besley and Case, 1993](#); [Foster and Rosenzweig, 1995](#); [Munshi, 2004](#); [Bandiera and Rasul, 2006](#); [Conley and Udry, 2010](#); [Dupas, 2014](#)). Its most familiar regularity is the S-shaped diffusion curve: take-up initially accelerates and later decelerates ([Rogers, 2003](#); [Young, 2009](#)). Figure 1 presents the classic hybrid-corn series reported by [Ryan and Gross \(1943\)](#).¹ Much of the diffusion literature obtains this shape from myopic threshold adoption, contagion, or exogenous opportunities. Here it arises in an equilibrium with fully forward-looking agents who strategically wait for information.

This paper asks how a principal should release supply when agents are forward-looking, differ in their value for a good innovation, and learn from a public bad-news process whose intensity rises with cumulative adoption. The principal cannot force adoption and cannot use transfers. Instead, she commits to a dated cumulative-supply policy and chooses it to minimize the time needed to reach an adoption target.²

¹The figure motivates the diffusion pattern rather than testing the model’s mechanism. The paper asks whether strategic waiting and heterogeneous payoffs can generate the shape without imposing myopia or an exogenous contagion process.

²The completion-time objective is a reduced-form representation of rollout programs evaluated by how quickly they reach a prespecified coverage or penetration threshold. It does not treat speed as a complete welfare criterion; welfare and distributional implications are discussed in Section 10.

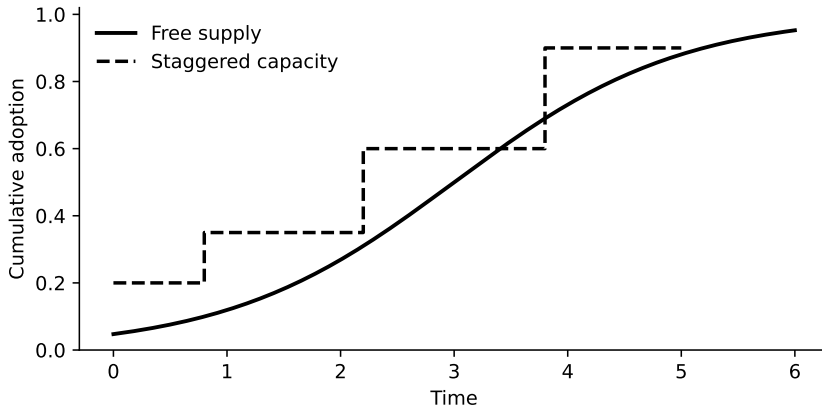


Figure 2: Free availability and staggered capacity. Dated scarcity changes the option value of waiting and can induce earlier adoption even when total supply is unchanged.

The design problem turns on the interaction of *screening* and *learning*. Fine staggering separates nearby valuations because high-value agents accept early opportunities while lower-value agents wait. Pooling gives up some of that screening precision, but it also creates a discrete increase in the stock producing information. Rationing adds a third margin. An early opportunity may attract an intermediate type by offering a chance of prompt service, yet remain unattractive to a higher type when rejection leads to a sufficiently delayed continuation. Consequently, the principal designs not only release dates, but also lotteries over service dates generated by applications, acceptance probabilities, and reapplication after rejection.

Figure 2 contrasts the benchmark and the policy logic. Under free supply, the ability to adopt later sustains a gradual equilibrium path. A dated-capacity plan removes some future options, brings experimentation forward, and changes the subsequent speed of learning. Scarcity is useful not because the principal likes withholding units per se, but because the future availability of units is the source of informational free-riding.

The analysis proceeds from the decentralized benchmark to finite operational design and then to a continuous-value direct-scheduling problem. The two policy classes are deliberately distinct. The finite model allows applications, anonymous rationing, rejection, and reapplication; the continuum problem assigns almost every value one service date and isolates the geometry of smooth separation and pooled rollout waves.

Free supply and the S-shape. I begin with unrestricted availability, under which every applicant is served. When one payoff type adopts gradually, indifference between neighboring dates pins down a one-dimensional ridge and an explicit growth law, so adoption rises convexly within that class. Heterogeneity then links these convex segments through waiting intervals. As adoption moves down the payoff distribution, the remaining agents gain less from successful adoption, which eventually slows diffusion. In the uniform case, the aggregate path has a unique inflection point under explicit endpoint conditions: adoption first accelerates and later decelerates. Thus strategic delay and heterogeneous values alone can generate the familiar S-shape.

Finite rollout design. Two preliminary results remove inessential complications. For any fixed bound on release dates, the selected-equilibrium problem has an optimum. In addition, unused inventory can be postponed until the date at which it is actually allocated. Hence every retained release may be taken to exhaust immediately, and the complexity of the finite problem comes from timing and rationing rather than idle stock.

The role of scarcity is sharpest with one value. Releasing exactly the target mass at date zero makes delay unstable: waiting now risks losing access, without creating a private informational advantage once everyone else can wait as well. The target is therefore reached immediately. With two target-relevant values, one initial and one terminal exhausted batch still attain the unrestricted optimum, although for a different reason. The low type cannot enter until enough exposure has accumulated, while the high type must not prefer the terminal opportunity. The larger of these two lower bounds determines completion. This result is global: it allows arbitrary finite calendars, repeated applications, and splitting within a type.

The two-value solution might suggest a general one-batch-per-value principle, but the three-value case shows why that intuition fails. A small early rationed opportunity for the intermediate type can preserve the highest type's incentive to adopt immediately. A later clearing release then serves rejected intermediate agents before the terminal date, raises the learning stock, and shortens completion. I verify the comparison against the entire simple class of plans with at most three immediately exhausted releases. In the counterexample economy, the best such plan completes at 4.1700003382, whereas a feasible four-release plan completes at 4.1402437122.³ All four releases exhaust, and the comparison relies only on on-path incentive conditions. The verified mechanism gives intermediate agents a nondegenerate rejection-contingent service lottery and establishes that this instrument can matter in a finite economy.

Additional releases are not always useful, however. If rejecting an early opportunity preserves certain service at a later clearing date, and early service is no better than that continuation, the early rationed offer cannot separate the residual block. It can then be removed without changing the realized path. A primitive mass-gap inequality, expressed in values, previously served mass, and the learning and discounting parameters, guarantees this continuation-value dominance.

Continuous-value direct scheduling and distributional shape. I next study a complementary policy class in which the principal assigns almost every value one certain service date. This is an analytical benchmark rather than a claim that operational rationing becomes infeasible or disappears under refinement. Single crossing orders service dates by payoff, and every continuously separating portion follows

$$B^*(v) = \frac{rv}{r + \beta[1 - F(v)]}.$$

The Appendix strengthens this local observation into a global decomposition. The prior and target determine unique compulsory boundary contacts m_0 and m_1 . Every finite-completion ordered deterministic schedule serves at least the upper mass m_0 at its first date and at least the

³The Appendix enumerates all binding patterns for three-release plans, reduces the competitive family to a constrained three-variable problem, and verifies its global minimum. The displayed precision records that calculation; no economic claim depends on the last reported digits.

lower target mass $\bar{M} - m_1$ at its terminal date; between those cores it consists of smooth ridge passage and a finite or countable family of pooled excursions. A boundary-attached excursion may enlarge the observed initial or terminal batch, so the compulsory cores need not equal the full boundary jumps for a general distribution.

This decomposition yields a sharp characterization. If the density is positive and nonincreasing, every nondegenerate excursion takes strictly longer than smooth passage through the same public states. Boundary-attached excursions are then excluded as well, and the unique deterministic optimum is exactly an initial batch of mass m_0 , a smooth separating phase, and a terminal batch of mass $\bar{M} - m_1$. If instead density rises sufficiently rapidly on an active region, small reflected pools are locally faster. A global replacement condition embeds that local gain in a deterministic optimum containing a wave, while a stronger no-fragmentation condition yields a wave covering the whole region.

Within the schedule-valued class, the rollout pattern therefore need not be uniformly smooth or uniformly pooled: it may combine boundary batches, smooth separation, interior pooled waves, and later smooth phases. The scope relative to relaxed lottery-valued continuum service is stated once in Remark 1.

1.1 Contribution relative to the literature

The paper sits at the intersection of strategic experimentation, innovation diffusion, dynamic pricing and recommendation, and artificial scarcity.

Strategic experimentation and irreversible adoption. [Chamley and Gale \(1994\)](#) and related models identify option-value-driven delay when actions are irreversible and information is generated by others. [Keller, Rady, and Cripps \(2005\)](#) and [Keller and Rady \(2015\)](#) develop conclusive-signal experimentation environments closely related to the bad-news technology used here. [Frick and Ishii \(2024\)](#) study forward-looking social learners in a decentralized adoption game and characterize waiting, partial adoption, and full adoption. Their equilibrium is the natural free-supply benchmark for this paper. My contribution is normative: a principal controls the timing of quantity and chooses a rollout to minimize completion time. The one-value scarcity result eliminates the within-class gradual phase, while the heterogeneous design problem introduces screening and rationing margins absent from the homogeneous decentralized environment.

Pricing, recommendation, and collective learning. [Bonatti \(2011\)](#) studies monopoly pricing and learning for a durable good with reversible use. [Che and Hörner \(2018\)](#) study recommendation systems for short-lived consumers who cannot strategically delay. [Laiho, Murto, and Salmi \(2025\)](#) analyze gradual learning from incremental actions, and related work by Laiho and Salmi studies dynamic pricing when adoption changes information. These papers share the idea that a large player can internalize an informational externality. The instrument and objective here are different: dated capacity rather than price or recommendation, and speed to a target rather than profit or welfare. Long-lived agents can reject, wait, and reapply, so the mechanism must control continuation lotteries after rejection.

Scarcity as a design instrument. Artificial scarcity also appears in [DeGraba \(1995\)](#), [Nocke and Peitz \(2007\)](#), and [Möller and Watanabe \(2010\)](#), generally without endogenous public learning. [Parakhonyak and Vikander \(2023\)](#) use scarcity to change the informational content of observed actions in a herding environment. Their mechanism belongs to the Bikhchandani–Hirshleifer–Welch tradition ([Banerjee, 1992](#); [Bikhchandani, Hirshleifer, and Welch, 1992](#); [Smith and Sørensen, 2000](#); [Eyster and Rabin, 2014](#)); scarcity coarsens or reshapes the public signal extracted from others’ choices. Here information arrives directly through a public experimentation technology, and scarcity works by limiting the option to postpone one’s own experimentation. The two mechanisms are complementary but distinct.

Diffusion and networks. [Rogers \(2003\)](#) and [Young \(2009\)](#) survey innovation diffusion across economics, marketing, and sociology. Social learning has been documented in agriculture and health in the papers cited above. Network models such as [Mossel, Sly, and Tamuz \(2015\)](#) and [Akbarpour and Jackson \(2018\)](#) emphasize who observes whom. The present model abstracts from network topology to isolate strategic timing with a public signal. Its S-shape result therefore complements network and contagion explanations rather than replacing them.

Mechanism-design interpretation. The finite policy is a dynamic mechanism implemented with anonymous capacity. In general a type’s allocation may be a distribution over service dates, and the counterexample shows that rejection-contingent continuation can itself be a design instrument. This recursive screening perspective is related to dynamic mechanism design, although the absence of transfers and the endogenous public-learning technology distinguish the present environment from the canonical framework of [Pavan, Segal, and Toikka \(2014\)](#). The continuum section instead studies direct service schedules over a continuous value distribution. Single crossing orders those schedules, while service continues to affect the public state through learning.

Roadmap. Section 2 presents the operational model and equilibrium. Section 3 characterizes free supply and the S-shape. Section 4 establishes fixed-release existence and the immediate-exhaustion representation. Section 5 solves the one- and two-type design problems, and Section 6 presents the three-type boundary result. Section 7 gives conditions under which rationing cannot separate a residual block. Section 8 formulates and characterizes the continuous-value direct-scheduling problem and records a one-directional finite approximation result. Section 9 proves the batch–smooth–batch result for nonincreasing densities and the pooling results for concentrated distributions. Section 10 discusses applications, commitment, and interpretation. Proofs and auxiliary calculations are in the Appendix.

2 Environment

2.1 Agents, values, and payoffs

A unit mass of agents is indexed by $i \in I = [0, 1]$. In the finite model, values belong to

$$v_1 > v_2 > \cdots > v_N > 0,$$

with type masses $q_n > 0$ summing to one. Later, values are distributed according to a cdf F with compact support $[\underline{v}, \bar{v}]$.

There is a persistent state $\omega \in \{g, b\}$. Agents and the principal share prior $\mu_0 = \mathbb{P}(\omega = g) \in (0, 1)$. Successful adoption is irreversible and yields v_i in the good state and -1 in the bad state. An agent who has not adopted receives zero flow payoff. All agents discount at rate $r > 0$.

Let μ_t be the posterior probability of the good state conditional on no bad signal. It is convenient to work with bad-state odds

$$B_t := \frac{1 - \mu_t}{\mu_t}.$$

Multiplying expected utility by the common positive factor $1/\mu_t$ leaves all comparisons unchanged. The normalized time-zero payoff from certain service at date t is therefore

$$g(v, t) := e^{-rt}(v - B_t). \quad (1)$$

4

2.2 Public learning

Bad news arrives only in the bad state. Conditional on $\omega = b$, its instantaneous intensity is βM_t , where M_t is cumulative successful adoption and $\beta > 0$. A bad signal reveals the state and ends adoption. Along the no-signal path,

$$B_t = B_a \exp \left\{ -\beta \int_a^t M_s ds \right\}, \quad t \geq a. \quad (2)$$

Thus adoption is productive experimentation: a larger stock generates faster information. An atom at date t does not change B_t instantaneously, but it increases the rate at which beliefs move immediately afterward.

⁴Heterogeneity in v also has an equivalent heterogeneous-prior interpretation. Suppose agents share a common good-state payoff $\bar{v}$ but type i has initial bad-state odds B_{0i} , with all types facing the same likelihood process. Multiplying type i 's payoff by the positive constant B_0/B_{0i} yields the common-prior formulation with $v_i = (B_0/B_{0i})\bar{v}$. Thus higher v_i corresponds to greater initial optimism. The transformation preserves individual choices, equilibrium paths, and the principal's completion-time objective, though not an unweighted utilitarian welfare sum without a corresponding change in welfare weights.

2.3 Supply, applications, and rationing

The principal commits at date zero to a right-continuous nondecreasing cumulative supply path S . In the finite operational model, S has finitely many jumps. At date t , let A_t be applicant mass and M_t^- prior successful adoption. Anonymous acceptance is

$$Q_t = \begin{cases} 0, & M_t^- = S_t, \\ 1, & M_t^- < S_t \text{ and } A_t \leq S_t - M_t^-, \\ \frac{S_t - M_t^-}{A_t}, & A_t > S_t - M_t^- > 0. \end{cases} \quad (3)$$

Unsuccessful applicants remain eligible. Rationing outcomes are private, while the aggregate adoption stock and public signal are observed.

Agents of one value need not choose the same realized application date. Under free supply, a value class can adopt over an interval. Under scarcity, a split type can be represented either by asymmetric one-shot dates or by common application followed by random rejection and reapplication. I allow behavioral mixed strategies after every private history. Atomlessness makes the induced aggregate paths deterministic conditional on the public history.⁵

2.4 Continuation values and equilibrium

Fix conjectured aggregate paths (M, A) and the induced acceptance path Q . A pure continuation plan α is a sequence of future application dates, contingent on the agent's private rejection history. If T^α is the realized set of attempts after date t , the normalized continuation value of type v is

$$W_t^\alpha(v) = \sum_{s \in T^\alpha, s > t} Q_s \prod_{k \in T^\alpha: t < k < s} (1 - Q_k) e^{-r(s-t)} \left[v - B_t e^{-\beta \int_t^s M_u du} \right]. \quad (4)$$

Behavioral values integrate (4) over private randomization. Let $W_t(v)$ be the supremum over continuation plans.

Definition 1 (Equilibrium). Given a supply path S , an equilibrium consists of behavioral application strategies and aggregate paths (M, A) such that: (i) after every private history each agent maximizes continuation value; (ii) every date used with positive probability is optimal for that type at that history; (iii) A and M are generated by the strategies and (3); and (iv) $M_t \leq S_t$ for all t . When several equilibria exist, the principal selects her preferred equilibrium.

The equilibrium-selection convention matters at weak inequalities. It is the same convention used in the one-value unraveling result and in the three-value construction, where type 2 and the terminal type are indifferent at selected dates.⁶

⁵Individual rationing outcomes remain private and random. Determinism refers only to aggregate applicant and service masses, which obey the corresponding acceptance probabilities almost surely in the atomless population.

⁶Principal-preferred selection is used to choose among equilibria generated by weak inequalities; it is not a refinement requiring all indifferent agents to apply at once. Such a rule would eliminate the mixed stopping that supports gradual free-supply adoption.

2.5 The principal

The principal fixes a target $\bar{M} \in (0, 1)$ and minimizes

$$T(S) := \inf\{t : M_t \geq \bar{M}\}$$

along the selected no-signal equilibrium path. In the finite-type results, the target is high enough to require positive service from every listed type; in particular, the two-type analysis assumes $q_H < \bar{M} < 1$. In the continuum sections, $\bar{M} < 1$ instead selects an upper portion of the value distribution. This distinction keeps the finite operational results separate from the continuous-value direct-scheduling problem.

A release is *immediately exhausted* if all new capacity is allocated at its release date. A plan is *K-release* if it contains at most K positive jumps. In an N -type economy, call a plan *simple* if it has at most N positive releases and every release is immediately exhausted. The old conjecture that some optimal finite plan is always simple is rejected below.

Lemma 1 (Upper-block willingness). *Fix a public and private history and any continuation strategy after rejection. The gain from applying at a positive-capacity date rather than following that continuation is strictly increasing in v . Hence willing residual agents form an upper block, with at most one indifferent cutoff type.*

The lemma orders successful-adoption distributions, not instantaneous actions after some types have already left the market. It also does not imply one service date per type: a cutoff type can be rationed, rejected, and served later.

3 Free supply and endogenous S-shapes

Free supply means $Q_t = 1$ whenever an agent applies. An individual therefore applies at most once, although a value class can be spread across dates through within-type mixing.

3.1 One value and the free-supply ridge

Let all residual agents have value v . Their payoff from certain service is $g_v(t) = e^{-rt}(v - B_t)$. On an interior interval of gradual adoption, nearby dates must be equally attractive, so $\dot{g}_v(t) = 0$. Using (2),

$$0 = e^{-rt} [\beta M_t B_t - r(v - B_t)].$$

The indifference ridge is

$$B_t = \frac{rv}{r + \beta M_t}. \tag{5}$$

Differentiating and substituting $\dot{B}_t = -\beta M_t B_t$ gives

$$\dot{M}_t = M_t(r + \beta M_t). \tag{6}$$

Thus cumulative adoption rises strictly and convexly while one class is active. The duration required to move from stock m_a to m_b on the ridge is

$$\Delta(m_a, m_b) = \int_{m_a}^{m_b} \frac{dm}{m(r + \beta m)} = \frac{1}{r} \log \left[\frac{m_b(r + \beta m_a)}{m_a(r + \beta m_b)} \right]. \quad (7)$$

Proposition 1 (One-class free-supply equilibrium). *If $v > B_0$, the free-supply equilibrium has the unique aggregate path obtained by jumping, when necessary, to the smallest feasible stock on (5) and then following (6) until the class is exhausted. If $v < B_0$, adoption never begins. At equality, both immediate and no-adoption equilibria are possible under weak tie selection.*

The convex path reflects strategic rather than technological acceleration. Early in the rollout, the stock generating information is small, so a substantial change in belief requires time. As adoption grows, learning speeds up, which makes still earlier adoption attractive to the remaining agents.

3.2 Several finite values

With $v_1 > \dots > v_N$, the free-supply path alternates between class-specific ridge segments and waiting intervals. A higher class exhausts before a lower class enters by single crossing. When type n finishes, the stock is fixed until belief reaches the entry ridge of type $n + 1$. The resulting path is piecewise convex with flat segments.

This structure clarifies why a finite-type path need not be smooth and why the model must permit within-type mixing. A claim that all agents of one type adopt at one date would contradict the free-supply benchmark. The relevant deterministic object is the aggregate path, which may be generated by asymmetric pure dates or by a common mixed stopping rule.

3.3 An atomless benchmark and the S-shape

Let values be continuously distributed with cdf F . If v_t is the marginal active value and $M_t = 1 - F(v_t)$ is the mass already served from the upper tail, local indifference gives

$$v_t = B_t \left(1 + \frac{\beta M_t}{r} \right), \quad M_t = 1 - F(v_t). \quad (8)$$

Equation (8) implicitly determines the aggregate free-supply path. It reduces the decentralized equilibrium to a scalar cutoff.

For a uniform distribution, differentiation gives a transparent curvature test. Early adoption accelerates because the learning stock is growing; late adoption decelerates because the remaining types become increasingly reluctant and the lower boundary of support approaches.

Proposition 2 (Uniform diffusion curves). *Let $v \sim U[\underline{v}, \bar{v}]$. Suppose the initial marginal type is interior and value dispersion is large enough that adoption traverses a nondegenerate interior range. Then the free-supply equilibrium is strictly increasing and, under an explicit prior threshold, has a unique inflection point: it is initially convex and eventually concave.*

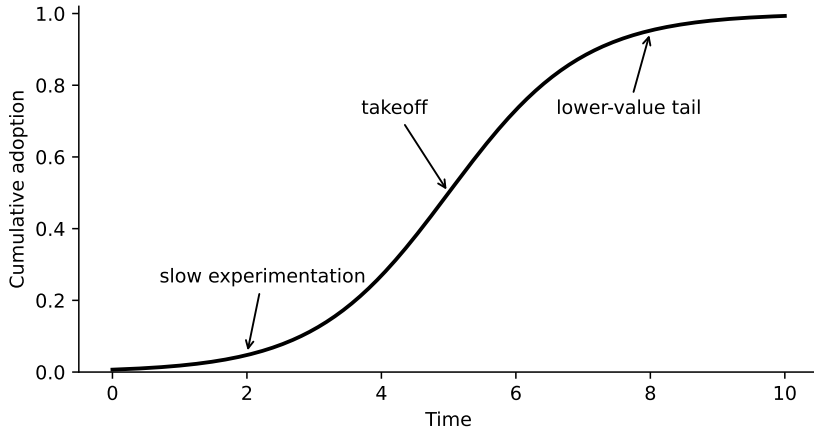


Figure 3: A strategic free-supply S-curve. Adoption is initially slow because little information is being generated. As adoption accumulates, learning accelerates and adoption speeds up. Later, the agents who have not yet adopted stand to gain less from successful adoption, so diffusion slows again.

Figure 3 illustrates the resulting strategic S-shape. The uniform case is useful because the inflection condition is closed form. The economic force is more general: a positive density over an active interval and sufficient dispersion can combine an accelerating learning stock with a progressively lower marginal value. The S-shape is a benchmark implication, not the main optimal-policy theorem.

4 Finite-plan normalization

Before solving the design problem, it is useful to remove two inessential complications. First, for any fixed upper bound on the number of release dates, the selected-equilibrium problem is well posed. Second, unused inventory is never needed: an outcome can be represented by postponing every unused unit until the date at which it is actually allocated. The substantive complexity studied below therefore comes from the timing and rationing of exhausted releases, not from idle stock.

4.1 Fixed-release existence

The failure of an N -release bound does not imply that finite optimization is ill posed. Fix K . Release dates, realized service masses, type-specific application shares, and rejection-contingent continuation values give a finite-dimensional formulation. Weakly ordered dates and zero-capacity dummy releases make the parameter set compact after imposing any feasible completion-time upper bound, while sequential optimality is represented by closed complementarity inequalities.⁷

Proposition 3 (Fixed- K existence). *For every finite economy and fixed upper bound K on releases, the principal's selected-equilibrium problem has an optimal K -release plan.*

⁷Writing realized service directly, rather than treating the acceptance ratio in (3) as a primitive at zero demand, removes the apparent discontinuity. The Appendix gives the compact parameterization.

Thus finite-calendar optimization is well posed for every fixed K . The number of release dates remains an independent design margin.

4.2 Immediate exhaustion survives

Suppose a release leaves inventory unused until a later date. Postpone the unused increment to the first date at which it is actually allocated. Earlier acceptance opportunities and all realized service are unchanged. Repeating this operation removes slack inventory.

Proposition 4 (Immediate-exhaustion representation). *For every optimal outcome among plans with at most K releases, there is an optimal representative with the same adoption path in which every positive retained release is immediately exhausted.*

This is the structural statement that survives the three-value correction. The principal may need more releases than values, but she never needs to leave already released units idle in order to implement a given optimal path.⁸

5 Scarcity with one and two values

The preceding normalization lets us focus directly on the economics of scarce timing. Every positive release below may be treated as immediately exhausted, and fixed-calendar optima exist. The remaining question is how many exhausted opportunities are useful and which values should face them.

5.1 One value: scarcity eliminates within-class delay

Suppose the unique target-relevant value satisfies $v > B_0$ and $\bar{M} < 1$. Release exactly $\bar{M}$ units at date zero. If all agents wait until a positive date, any one agent prefers applying slightly earlier because beliefs are unchanged before adoption begins and discounting is smaller. If adoption is gradual while inventory remains, an agent assigned later prefers applying just before the assigned date. If the target is reached at a positive date through a final rationed rush, an applicant prefers an arbitrarily earlier chance at the same scarce stock. These deviations unravel every delayed equilibrium selected by the principal.

Theorem 1 (One-value optimum). *With one target-relevant value $v > B_0$ and target $\bar{M} < 1$, releasing exactly $\bar{M}$ units at date zero reaches the target immediately. No policy can do better.*

The result illustrates the core force of scarcity in its purest form. Free supply creates an unconditional option to wait. A capacity cap turns delay into a risk of exclusion and eliminates same-type free-riding.

⁸The operation postpones slack inventory to the date at which it is first allocated. It does not merge distinct exhausted releases and therefore does not imply any bound on their number.

5.2 Two values: exposure and incentive lower bounds

Let $v_H > B_0 > v_L$, let the high-type mass be q_H , and suppose $q_H < \bar{M} < 1$, so completion requires some low-type service. An optimal policy must satisfy two independent lower bounds.

First, even if all high agents adopt immediately, the low type cannot become willing before exposure drives belief from B_0 to v_L :

$$T \geq T_E := \frac{1}{\beta q_H} \log \frac{B_0}{v_L}. \quad (9)$$

Second, a high type served at zero must not prefer waiting for certain terminal service with the low type. Since terminal participation is fastest at $B_T = v_L$,

$$v_H - B_0 \geq e^{-rT}(v_H - v_L), \quad (10)$$

which is equivalent to

$$T \geq T_H := \frac{1}{r} \log \frac{v_H - v_L}{v_H - B_0}. \quad (11)$$

The principal can attain the larger of the two bounds. Let

$$T^* = \max\{T_E, T_H\}, \quad m_0^* = \frac{1}{\beta T^*} \log \frac{B_0}{v_L}.$$

Release m_0^* at zero and the residual target mass at T^* . If $m_0^* < q_H$, all high agents may apply at zero and be rationed, or a deterministic mass m_0^* may be assigned to zero. Rejected or postponed high agents join the terminal batch. Indifference makes the two representations equivalent.

Theorem 2 (Unrestricted two-value optimum). *Suppose $v_H > B_0 > v_L$ and $q_H < \bar{M} < 1$. The minimum completion time over all finite supply plans is*

$$T^* = \max \left\{ \frac{1}{r} \log \frac{v_H - v_L}{v_H - B_0}, \frac{1}{\beta q_H} \log \frac{B_0}{v_L} \right\}.$$

An optimal plan uses one initial and one terminal immediately exhausted release, with initial service mass m_0^ as above and certain terminal service for every residual target agent.*

Figure 4 displays the two regimes. The theorem is the proved positive boundary of the finite model. It does not rely on a universal compression theorem. With only one higher type to discipline and one lower entry condition, a single initial lottery and a terminal clearing opportunity span the relevant margins.

Comparative statics. Completion is weakly decreasing in β and q_H . A higher discount rate relaxes the high-type waiting constraint but also changes the free-supply ridge; in the two-type formula its direct effect is to reduce T_H . A more optimistic prior lowers the immediate payoff loss of high adoption but raises the exposure required to convince the low type, so the sign can differ across the two regimes.

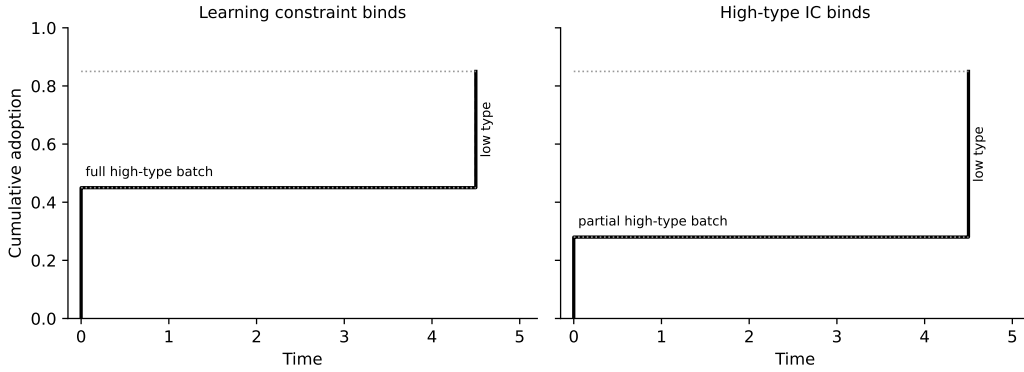


Figure 4: Optimal adoption paths with two values. Left: when learning is the binding constraint, all high-value agents adopt at date zero and the low-value agents enter at the earliest feasible date; the principal attains the first-best learning bound. Right: when the high type’s incentive to wait is binding, only part of the high type is served at date zero, and the residual high-type mass joins the low type at the terminal batch.

5.3 Scarcity relative to free supply

The one- and two-value results are not isolated curiosities. They reveal a general reason why dated capacity can strictly improve on unrestricted availability. Under free supply, the last target-relevant type enters only when its continuation value from waiting has fallen to its current service payoff. A small amount of scarcity at that entry margin changes the comparison discontinuously: an agent who postpones application may lose the unit rather than merely receive it later. The principal can therefore advance at least part of the marginal type without reducing the amount of prior experimentation.

The argument is easiest to see at a candidate free-supply completion date T^F . Let v_L be the marginal target-relevant value. Immediately before T^F , an active v_L agent is indifferent between service and continued waiting. Replace free availability near T^F by a terminal capacity equal to the residual target mass. If the applicant mass strictly exceeds that capacity, the batch rations and exhausts. Applying slightly earlier becomes strictly attractive because it increases the probability of receiving the scarce unit. If applicant mass equals capacity, an arbitrarily small reduction in terminal capacity creates the same force. The resulting extra adoption raises the learning stock before the original completion date.

Proposition 5 (Strict value of scarcity). *Suppose the free-supply equilibrium reaches a target $\bar{M} < 1$ at a positive finite date and the last target-relevant type has positive residual mass at entry. Then there exists a finite dated-capacity policy whose selected equilibrium reaches the target strictly earlier.*

The proposition isolates the economic force: eliminating an unconditional future option can improve completion even before asking whether the optimal policy is simple or whether one perturbation works uniformly across equilibria.

Private and social welfare. The principal’s objective is completion time, not utilitarian welfare. Scarcity can nevertheless improve adopter welfare in some regions because it brings information production forward and can reduce the waiting costs borne by later agents. It can also hurt agents who are rationed or induced to experiment earlier. The paper therefore does not treat faster completion as automatically Pareto improving. The mechanism-design results characterize the speed frontier; welfare comparisons depend on the application and on whether the target itself generates external benefits not internalized by agents.

6 Three values: repeated rationing defeats the batch-count bound

The two-value theorem suggests that one exhausted release per value class might suffice. The conjecture fails with three values. The reason is not idle inventory: by proposition 4, every positive release can be taken to exhaust immediately. The new instrument is instead a *sequence of rationed opportunities* for the same intermediate type.

In the counterexample, the top type is served immediately. The intermediate type then applies at two interior dates: a small fraction is accepted at the first date, while rejected agents remain eligible and are cleared at the second. The lowest target-relevant type enters only at the terminal date. The first intermediate opportunity is deliberately small because a larger early probability of service would attract the top type away from date zero. The second opportunity is deliberately large because postponing most of the intermediate type until the terminal date would sacrifice learning. Thus the two interior releases perform different incentive tasks and cannot generally be collapsed into one.

Theorem 3 (Simple-plan optimality fails with three values). *There exists a three-value economy in which a feasible plan with four immediately exhausted releases completes strictly earlier than every immediately exhausted plan with at most three positive release dates. Consequently, the claim that every finite N -type economy has an optimal plan with at most N immediately exhausted releases is false already for $N = 3$.*

Figure 5 summarizes the mechanism. The comparison in theorem 3 is global within the simple class, not a comparison with one guessed three-date policy. The appendix enumerates the possible upper-block applicant modes, reduces the best-simple-plan problem to a finite comparison of binding cases, and verifies a strict completion-time gap. The strict gap persists under sufficiently small perturbations of the primitives.

The lesson is narrow but important. Simple-plan optimality is not universal, and repeated eligibility can be economically substantive. The counterexample’s operational equilibrium genuinely uses rejection-contingent rationing: all intermediate agents apply at the first interior date, only a fraction are served, and rejected agents remain eligible for the later clearing release. Replacing that lottery by an ex ante partition into certain dates would change the anonymous-rationing deviation faced by the top type and is not part of the proof. This is also why the continuous-value analysis below is posed as a direct-scheduling problem rather than derived as the unrestricted limit of the operational problem.

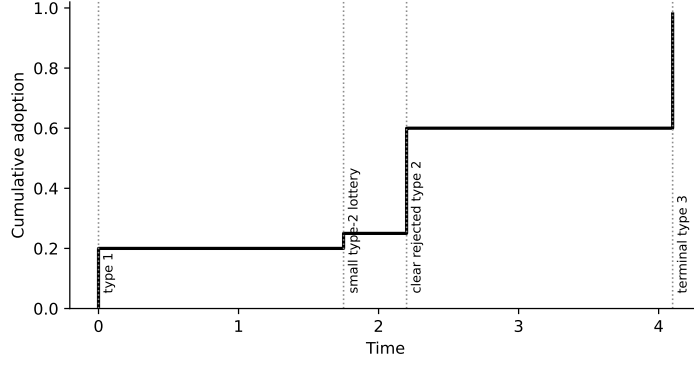


Figure 5: The four-release plan. Type 1 is served immediately. Type 2 receives a small early rationed opportunity and, after rejection, a later clearing opportunity. The final release serves the marginal lower type. The full numerical verification, sequential incentive check, and global comparison with the simple class are in the Appendix.

7 When should types be pooled?

The three-value construction shows that rationing can profitably separate values. This section identifies the opposite case: circumstances under which an earlier rationed opportunity cannot draw a residual higher type away from a later clearing batch.

The distinction matters because rejection is valuable only when it preserves a sufficiently attractive continuation. At a terminal batch this is immediate: an agent who rejects an earlier opportunity remains eligible for final clearing. At an interior clearing date, the same conclusion requires an explicit continuation guarantee.

7.1 Pooling at a protected clearing date

Consider a public history at which the residual population contains the consecutive block

$$v_k > v_{k+1} > \dots > v_j.$$

A date T is a *protected clearing date* for this block if:

- (i) every member of the block who remains unserved immediately before T receives certain service at T ;
- (ii) rejecting any opportunity at a date $t < T$ preserves eligibility for that service; and
- (iii) an opportunity rejected by every member of the block produces no adoption and hence does not change the public state or the continuation faced by other agents.

The third condition is automatic for an offer that receives no applications: deleting it leaves the realized adoption and belief paths unchanged. The first two conditions are automatic when T is the terminal clearing batch. When T is interior, they must be verified from the continuation policy.

For a residual type v_i , let

$$g_i(t) := e^{-rt}(v_i - B_t)$$

denote its time-zero payoff from certain service at t , and let $W_i(t^+)$ denote its time-zero continuation value immediately after rejecting an opportunity at t .

Proposition 6 (Pooling by continuation-value dominance). *Fix a continuation with a protected clearing date T for the residual block $v_k, \dots, v_j$. Suppose that, for every $i = k, \dots, j$ and every feasible opportunity date $t < T$,*

$$g_i(t) \leq W_i(t^+). \quad (12)$$

Then there is a sequential equilibrium of this continuation in which every member of the block rejects every pre- T opportunity and the residual block is pooled at T .

Every pre- T opportunity directed only at this block can be deleted without changing the realized adoption path, the belief path, or the completion date. Consequently, if the original continuation is optimal, the continuation obtained by deleting those unused opportunities is also optimal.

Proof. Suppose type v_i applies at $t < T$ and is accepted with probability Q . Its payoff is

$$Qg_i(t) + (1 - Q)W_i(t^+).$$

Under (12), this is weakly below $W_i(t^+)$ for every $Q \in [0, 1]$. Rejection is therefore optimal. If the inequality is strict, rejection is the unique best response whenever $Q > 0$.

At T , service is certain. Since the lowest value v_j is willing to accept and every other member of the block has a higher value, upper-block willingness implies that all remaining types $v_k, \dots, v_j$ weakly prefer to apply. Thus the specified strategies are sequentially optimal.

Because all pre- T opportunities directed only at this block are rejected, they generate no adoption and no belief change. Condition (iii) in the definition of a protected clearing date therefore permits their deletion without altering the realized continuation. The completion date and the principal's payoff are unchanged. $\square$

The proposition gives a local test for a proposed clearing block. It asks whether rationing before that date can accomplish any separation. If even certain early service is no better than rejection and continuation, reducing the acceptance probability cannot make the early opportunity attractive.

7.2 A primitive mass-gap condition

The continuation-value test is tight but endogenous. A stronger condition can be stated using only values, the mass already served, and the learning and discounting parameters.

Suppose that the lowest type in the residual block first becomes willing to receive certain service at T , so that

$$B_T = v_j.$$

Suppose also that cumulative adoption satisfies $M_t \geq m$ for every $t < T$ along the relevant continuation.

Proposition 7 (Primitive mass-gap condition). *Let T be a protected clearing date for the residual block $v_k > \dots > v_j$, and suppose $B_T = v_j$ and $M_t \geq m$ for all $t < T$. If*

$$\beta m v_j \geq r(v_k - v_j), \quad (13)$$

then condition (12) holds for every type in the block. Hence the block can be pooled at T , and every earlier opportunity directed only at that block can be deleted without affecting completion time.

If types $1, \dots, k-1$ have already been completely served, it is sufficient that

$$\beta v_j \sum_{\ell < k} q_\ell \geq r(v_k - v_j). \quad (14)$$

Proof. For any residual type $v_i \leq v_k$,

$$g'_i(t) = e^{-rt} [(r + \beta M_t) B_t - r v_i].$$

Beliefs decline over time, so $B_t \geq B_T = v_j$ for every $t < T$. Together with $M_t \geq m$ and $v_i \leq v_k$, this gives

$$\begin{aligned} g'_i(t) &\geq e^{-rt} [(r + \beta m) v_j - r v_k] \\ &= e^{-rt} [\beta m v_j - r(v_k - v_j)] \geq 0. \end{aligned}$$

Thus $g_i(t) \leq g_i(T)$ for every $t < T$. Because T is protected, rejecting at t preserves certain service at T , and therefore

$$W_i(t^+) \geq g_i(T) \geq g_i(t).$$

Proposition 6 applies.

If types $1, \dots, k-1$ have already been served completely, cumulative adoption is at least $m = \sum_{\ell < k} q_\ell$, which yields (14). $\square$

Condition (13) compares two forces:

$$\underbrace{\beta m v_j}_{\text{learning generated by prior adoption}} \quad \text{and} \quad \underbrace{r(v_k - v_j)}_{\text{discounting cost across the value gap}}.$$

More prior adoption, faster learning, or a narrower residual value range makes pooling more likely. Because the condition evaluates learning at the lowest value and waiting costs at the highest value, checking the two endpoints is sufficient for every intermediate type.

The inequality is deliberately stronger than necessary. It requires the certain-service payoff of the highest residual type to be increasing throughout the entire interval before T . The continuation-value test permits pooling even when this payoff falls temporarily, provided rejection offers a sufficiently valuable continuation.

7.3 How to check the general condition

On an interval over which cumulative adoption is constant at M ,

$$g'_i(t) = e^{-rt}[(r + \beta M)B_t - rv_i].$$

Because B_t declines, the term in brackets crosses zero at most once. The certain-service payoff is therefore single-peaked on that interval. Its interior peak, when present, occurs at

$$B_t = \frac{rv_i}{r + \beta M}.$$

Consequently, verifying (12) does not generally require checking every date. On each constant-stock interval, it is enough to compare continuation value with the payoff at the ridge date and at the relevant endpoints.

The economic conclusion is simple. A residual block should be pooled when accumulated learning makes waiting for the lower type's clearing date at least as attractive as every earlier rationed opportunity. Separation can be useful only when some earlier service opportunity rises above the continuation protected by rejection.

8 Continuous-value direct scheduling

The finite operational model allows applications, anonymous rationing, rejection, and reapplication. I now study a complementary direct-scheduling problem over a continuous value distribution. A plan assigns almost every served value one service date, subject to the same learning dynamics, participation, and incentive constraints. The restriction is analytical rather than institutional: it excludes rejection-contingent service lotteries in order to isolate the global geometry of smooth separation and pooled rollout waves. The three-value counterexample shows why the operational and direct-scheduling problems need not coincide.

8.1 Ordered deterministic direct schedules and the separating ridge

An ordered deterministic direct schedule assigns every served upper-tail rank $m \in [0, \bar{M}]$ one certain date $\tau(m)$, with τ weakly increasing. The word “deterministic” concerns service conditional on value: it permits a positive mass of different values to share a date, but excludes a nondegenerate distribution over dates for a positive set of values. Let λ denote Lebesgue measure on the rank interval $[0, \bar{M}]$. Since rank is measured in population mass, the adoption and belief paths are

$$M_t = \lambda\{m \in [0, \bar{M}] : \tau(m) \leq t\}, \quad B_t = B_0 \exp \left\{ -\beta \int_0^t M_s ds \right\}.$$

Global direct incentive compatibility requires

$$e^{-r\tau(m)}\{\nu(m) - B_{\tau(m)}\} \geq e^{-r\tau(n)}\{\nu(m) - B_{\tau(n)}\} \quad \text{for all served } m, n. \quad (15)$$

The ridge and density analysis below is global within this ordered schedule-valued class.

On a smooth separating component, local IC makes the marginal type indifferent across neighboring service dates. Writing $\nu(m) = F^{-1}(1 - m)$, the same calculation as under free supply gives

$$B^*(m) = \frac{r\nu(m)}{r + \beta m}. \quad (16)$$

If $x^*(m) = -\log B^*(m)$, the smooth time density is

$$\tau'(m) = \frac{x^{*'}(m)}{\beta m}. \quad (17)$$

Thus smooth passage from m_a to m_b takes

$$\mathcal{S}(m_a, m_b) = \int_{m_a}^{m_b} \frac{x^{*'}(m)}{\beta m} dm. \quad (18)$$

The local ridge identity extends to the entire deterministic schedule.

Proposition 8 (Boundary cores and exact deterministic decomposition). *Suppose ν is positive and strictly decreasing, $\bar{M} \in (0, 1)$, and*

$$\nu(\bar{M}) < B_0 < \nu(0).$$

Let τ be any finite-completion ordered deterministic direct schedule satisfying global incentive compatibility, participation of the upper $\bar{M}$ mass, and exclusion below $\nu(\bar{M})$. Define $b(m) = r\nu(m)/(r + \beta m)$ and let the unique $0 < m_0 < m_1 < \bar{M}$ solve

$$b(m_0) = B_0, \quad b(m_1) = \nu(\bar{M}).$$

If $t_0 = \tau(0)$ and $T = \tau(\bar{M})$, then $\tau = t_0$ on $[0, m_0]$ and $\tau = T$ on $[m_1, \bar{M}]$. Between these compulsory boundary cores, every continuous increase is absolutely continuous ridge passage and every jump is an exact pooled excursion (d_j, p_j, h_j) . The excursion supports have disjoint interiors, there are at most countably many of them, and

$$T = t_0 + \mathcal{S}(m_0, m_1) + \sum_j \Gamma_F(d_j, p_j, h_j), \quad (19)$$

where

$$\Gamma_F(d, p, h) = \frac{1}{\beta} \int_d^h x^{*'}(m) \left(\frac{1}{p} - \frac{1}{m} \right) dm.$$

The observed initial batch may exceed m_0 only as the departure block of the first excursion, and the observed terminal batch may begin below m_1 only as the reentry block of the last excursion.

Thus m_0 and m_1 are unique for every admissible density, without imposing $f' \leq 0$. What remains distribution-dependent is the optimal excursion family. The proposition also rules out a singular-continuous service-time assignment and permits a finite or countable number of smooth and pooled components. In any optimum the removable idle time satisfies $t_0 = 0$.

8.2 Smooth rollout and pooled waves

A pooled wave replaces a segment of smooth separation by a discrete increase in adoption, a period of waiting while beliefs improve, and a second increase that returns the schedule to the ridge. Pooling need not be absorbing: smooth rollout can resume after the wave.

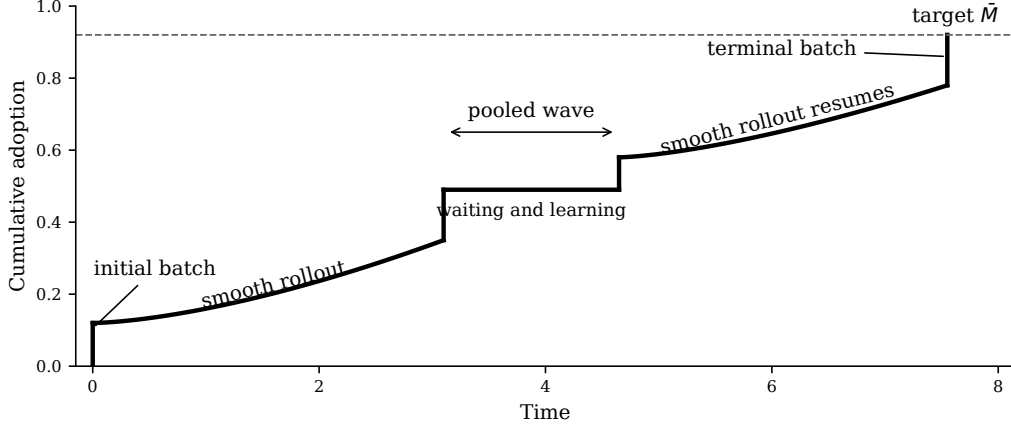


Figure 6: Smooth rollout and pooled waves within the ordered deterministic class. An interior wave produces a discrete increase in adoption, followed by a waiting interval while public beliefs adjust; smooth rollout may then resume. The final jump reaches the adoption target $\bar{M}$.

Figure 6 illustrates the possible components of an ordered service schedule. The next section compares the duration of a pooled excursion with smooth passage through the same public states.

8.3 Finite approximation and calendar refinement

The direct-scheduling problem is defined on its own. Two results connect it to finite economies without claiming convergence of unrestricted finite optima. First, strongly regular continuum schedules can be approximated by target-aligned finite direct menus. Second, terminal padding shows how a calendar with many release dates can become simple after an arbitrarily small refinement of terminal-only value classes.

8.3.1 Finite approximation of direct schedules

To connect this class to finite menus, consider ordered direct schedules rather than arbitrary application-game limits. A schedule is *strongly regular* if it has finitely many pools and smooth separating components, transverse endpoint contacts, adjacent-cutoff Jacobians bounded away from singularity, and uniform slack in every nonbinding global IC and lower-tail inequality.⁹

Proposition 9 (Finite approximation of strongly regular schedules). *Every strongly regular ordered direct schedule has target-aligned adjacent-cell finite direct menus with exact adjacent cutoff equalities, exhausted batches, global direct IC, and completion times converging to the continuum value.*

⁹The construction solves finitely many neighboring contact equations. The nonsingularity condition permits the ordinary finite-dimensional implicit-function theorem to be applied uniformly on a finite cover; no infinite-dimensional implicit-function result is used.

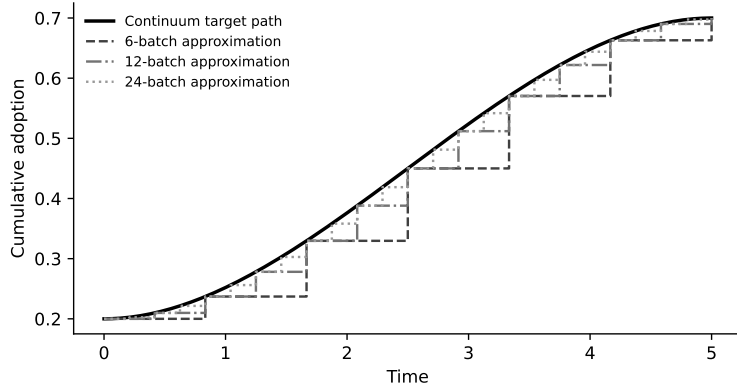


Figure 7: Finite exhausted-batch approximations of a regular continuum service schedule. Vanishing adjacent cells generate a smooth separating component; cells with nonvanishing mass generate pooled waves.

The proposition recovers the continuum schedule within finite direct menus. Operational implementation in the original application game additionally requires rejection-contingent deviations to be unprofitable.

Figure 7 illustrates this one-directional approximation. It shows that the direct-scheduling problem can be approached by finite direct menus; it does not assert that unrestricted finite optima converge to that benchmark.

8.3.2 Terminal padding and asymptotic calendar simplicity

Let an immediately exhausted finite equilibrium use dates $t_1 < \dots < t_K = T$, with beliefs $B_1 > \dots > B_K$ along the no-signal path. Values in the open interval (B_K, B_{K-1}) strictly prefer to make no preterminal application and to apply at T : every earlier application carries a positive chance of immediate service with negative flow payoff, while terminal service has positive payoff. This one-sided strictness permits exact type padding.

Proposition 10 (Terminal padding). *Suppose a finite immediately exhausted equilibrium has a positive mass of terminal-only applicants whose values lie in an open interval $I \subset (B_T, B_{K-1})$. For any integer $L \geq 1$, one may divide an arbitrarily small positive submass of those applicants among L distinct values in I , leaving every application mass, acceptance probability, adoption path, belief path, and completion time unchanged. Hence a K -release outcome can be represented in an arbitrarily nearby economy with at least K value classes and is simple there.*

Padding preserves the equilibrium outcome exactly. Along suitable terminal-rich refinements, the same construction can also make the calendar-count advantage vanish asymptotically. The precise limiting statement is recorded in the Appendix.

The counterexample and the padding result concern different margins. Additional dates can have strict value in a finite operational economy, while refinement can remove the calendar-count disadvantage of simple plans. Neither result identifies the direct-scheduling problem with the unrestricted operational limit.

9 Distributional forces: declining density and concentration

All optimality statements in this section concern the ordered schedule-valued class defined in Section 8.

9.1 Same-state pooled excursions

Consider a ridge-returning excursion indexed by a departure stock m_d , a pooled stock $p > m_d$, and a reentry stock $m_r > p$. The schedule releases $p - m_d$ units at departure, waits with adoption fixed at p , and then releases $m_r - p$ units at reentry. Let $a = \nu(p)$ and $k = r/(\beta p)$. Feasibility requires the departure and reentry payoff lines to cross at a and the belief law to connect the two ridge states. The Appendix shows that the resulting lines fit into the ordered payoff envelope and that every jump of an arbitrary globally incentive-compatible deterministic schedule has exactly this form.

Let $x^*(m) = -\log B^*(m)$. The excursion generates the same log-belief decline as smooth passage, but with adoption fixed at p , so its waiting time is

$$\Delta(m_d, p, m_r) = \int_{m_d}^{m_r} \frac{x^{*'}(m)}{\beta p} dm.$$

Define its excess duration

$$\Gamma_F(m_d, p, m_r) := \Delta(m_d, p, m_r) - \mathcal{S}(m_d, m_r). \quad (20)$$

A negative value means that pooling is faster.

9.2 Positive nonincreasing densities

Write f for the density. Since $\nu'(m) = -1/f(\nu(m))$,

$$x^{*'}(m) = \frac{\beta}{r + \beta m} + \frac{1}{\nu(m)f(\nu(m))}. \quad (21)$$

When f is nonincreasing in value, inverse-tail slopes have the convexity that makes every nondegenerate excursion slower than smooth passage.

Theorem 4 (Unique deterministic optimum under a nonincreasing density). *Suppose F has a positive continuously differentiable density f on the relevant support, f is nonincreasing, $\bar{M} \in (0, 1)$, and*

$$\nu(\bar{M}) < B_0 < \nu(0).$$

Define

$$b(m) = \frac{r\nu(m)}{r + \beta m}.$$

There are unique masses $0 < m_0 < m_1 < \bar{M}$ satisfying

$$b(m_0) = B_0, \quad b(m_1) = \nu(\bar{M}). \quad (22)$$

Among all finite-completion ordered deterministic direct schedules that serve the upper $\bar{M}$ mass and satisfy global IC, participation, and lower-tail exclusion, the unique aggregate optimum is

$$\tau^*(m) = \begin{cases} 0, & 0 \leq m \leq m_0, \\ \int_{m_0}^m \frac{x^{*'}(z)}{\beta z} dz, & m_0 < m < m_1, \\ \mathcal{S}(m_0, m_1), & m_1 \leq m \leq \bar{M}. \end{cases} \quad (23)$$

Thus the optimal deterministic rollout consists of an initial batch, smooth separation, and a terminal batch. Uniqueness is understood for the aggregate completed graph, apart from zero-mass cutoff conventions.

The proof has three parts. First, the payoff-envelope argument shows that continuous service lies on the ridge, rules out a singular-continuous service-time component, and turns every waiting interval into an actual departure batch, wait, and reentry batch. This yields the exact finite-or-countable decomposition in (19), including possible excursions attached to the two boundary contacts. Second, under $f' \leq 0$, every nondegenerate excursion term is strictly positive. This eliminates boundary-attached excursions as well as interior waves, so the compulsory cores become the full initial and terminal batches. Third, the Appendix verifies attainment directly: along the candidate ridge menu, each type's payoff is single-peaked at its assigned rank, while the date-zero and terminal lines are optimal for the two boundary blocks. The theorem therefore imposes neither ridge attachment nor a finite number of smooth regions; both follow from global IC and the belief law within the deterministic class.

Remark 1 (Deterministic and relaxed continuum values). Let $V^D(F)$ denote the minimum completion time over ordered schedule-valued direct allocations, and let $V^R(F)$ denote the relaxed operational value, which permits a nondegenerate service-time kernel for a positive set of values. Theorem 4 proves

$$V^D(F) = \mathcal{S}(m_0, m_1)$$

and uniquely identifies its aggregate optimizer. It does not identify $V^R(F)$. Whenever the canonical schedule is operationally implementable or can be approximated by operational plans, the relaxed operational value can be no larger than its completion time; whether it can be strictly smaller remains open.

Corollary 1 (Uniform benchmark). *For a uniform distribution satisfying the interior boundary conditions of Theorem 4, the unique optimal deterministic rollout is the initial-batch–smooth–terminal-batch schedule in (23).*

9.3 An explicit nonincreasing-density optimum

Figure 8 computes the optimal deterministic rollout for an explicit distribution. Let $s(z) = 3z^2 - 2z^3$ and let $f(v) = \tilde{f}(v)/1.162$ on $[0, 1]$, where

$$\tilde{f}(v) = \begin{cases} 1.55, & 0 \leq v \leq .18, \\ 1.55 - .40 s\left(\frac{v - .18}{.27}\right), & .18 < v < .45, \\ 1.15, & .45 \leq v \leq .62, \\ 1.15 - .60 s\left(\frac{v - .62}{.38}\right), & .62 < v \leq 1. \end{cases} \quad (24)$$

This density is positive, continuously differentiable, and nonincreasing. It is flat on $[0, .18]$ and $[.45, .62]$, and strictly decreasing on the two intervening regions.

Set $\beta = 1$, $r = .35$, and $\bar{M} = .93$. Choose the prior so that $m_0 = .08$ solves $B_0 = B^*(m_0)$:

$$B_0 = 0.6927178048.$$

Let m_1 solve $B^*(m_1) = \nu(\bar{M})$. For (24), the target and terminal contact both lie in the lower flat-density region, where $\nu(m) = (1 - m)/f_0$ and $f_0 = 1.55/1.162$. The terminal equality reduces to

$$r(1 - m_1) = (1 - \bar{M})(r + \beta m_1),$$

which gives $m_1 = .775$. The rollout jumps to m_0 at date zero, follows the smooth ridge from m_0 to m_1 , and completes the target with a terminal batch from m_1 to $\bar{M}$. The smooth phase lasts 8.2542833 time units. The open circles in the right panel correspond to $v = .62, .45, .18$, the points at which the density moves between flat and declining pieces.

Proposition 11 (Optimal deterministic rollout for the example). *For the primitives above, the unique optimal deterministic aggregate schedule is the initial-batch–smooth–terminal-batch path displayed in Figure 8.*

Proof. The density is positive and nonincreasing, and the stated prior and target satisfy the boundary conditions of Theorem 4. The boundary equations give $m_0 = .08$ and $m_1 = .775$, so (23) yields the displayed path. $\square$

The example shows that a nonincreasing density eliminates interior waves without making the rollout visually featureless: the distribution continues to shape the speed and curvature of smooth adoption.

9.4 Sufficient convexity creates a pooling force

Near a ridge contact point, let a reflected local excursion have scale ε . Tangency makes the linear and quadratic terms in (20) cancel. Under C^4 regularity,

$$\Gamma_F = K_F(m)\varepsilon^3 + O(\varepsilon^4), \quad (25)$$

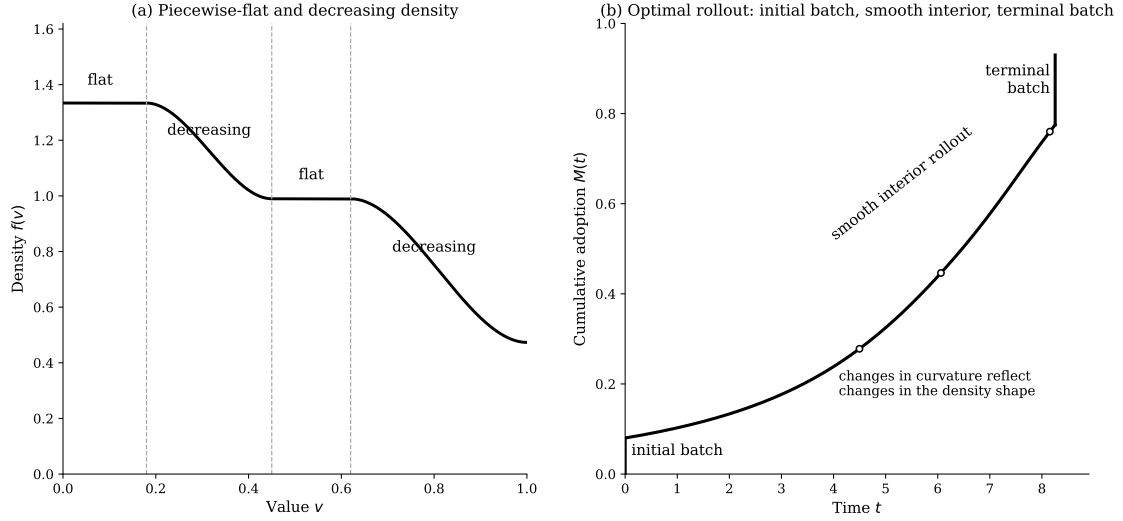


Figure 8: Density shape and the unique optimal deterministic rollout. Panel (a) plots the positive nonincreasing density in (24). Panel (b) shows the initial batch, smooth interior rollout, and terminal batch characterized by Theorem 4. Changes in the curvature of the smooth path reflect the density’s flat and strictly declining regions.

where

$$K_F(m) = \frac{2rf - \beta m \nu f' + 2\beta r \nu f^2 / (\beta m + r)}{3\beta^2 m^3 \nu^2 f^3}. \quad (26)$$

The denominator is positive. Hence a primitive sufficient condition for a local pooling gain is

$$\frac{f'(\nu(m))}{f(\nu(m))} > \frac{2r}{\beta m \nu(m)} + \frac{2rf(\nu(m))}{m(\beta m + r)}. \quad (27)$$

Because $F'' = f'$, condition (27) is a strong-convexity condition on the distribution over the active region: density rises fast enough that a large mass of nearby values can be advanced with little movement in the marginal value.

Theorem 5 (Sufficient convexity creates a local pooling gain). *Let C be a compact interior ridge core on which the reflected-contact map is regular, the fourth-order remainder in (25) is uniform, and $K_F \leq -\eta < 0$. Then every point of C supports sufficiently small exact reflected excursions with*

$$\Gamma_F < 0.$$

Thus smooth passage is locally slower than a pooled bridge between the same ridge endpoints.

Theorem 5 is primitive-facing but local. A faster same-state bridge need not be insertable into an arbitrary global schedule: advancing the continuation can tighten the waiting constraints of types served before the bridge. The following selection result states the additional global condition explicitly.

Proposition 12 (Global selection of a wave). *Let C satisfy Theorem 5. Suppose there is a negative-gain excursion e based in C with the following replacement property: for every optimal*

globally compatible ridge-contact family that omits e , the subfamily whose supports conflict with e can be replaced by e while preserving global incentive compatibility, and

$$\Gamma_F(e) \leq \sum_{a \in \mathcal{C}_e} \Gamma_F(a),$$

where $\mathcal{C}_e$ is the removed conflicting subfamily. Then some optimal deterministic rollout contains a nondegenerate wave intersecting C .

The replacement property is satisfied, for example, when the candidate lies at the beginning of the active region, or when all upstream waiting constraints have enough strict slack to absorb the induced acceleration of continuation values. It can also be verified directly from the finite set of binding envelope constraints in a proposed rollout pattern.

For a stronger whole-core statement, let $W(C)$ be the minimum gain among globally compatible single waves whose support contains C , and let $H(C)$ be the minimum total gain among globally compatible families that serve the same region without any one wave covering C .

Corollary 2 (Whole-core batching). *If the replacement property holds for outside components and*

$$W(C) \leq H(C),$$

then some optimal deterministic rollout contains a single wave whose support covers the whole core C .

The hierarchy is deliberate. Strong convexity of F yields a directly verifiable local force toward pooling. Global embeddability yields batching of at least part of the certified core. The no-fragmentation comparison $W(C) \leq H(C)$ yields a wave covering the whole core. None of these statements requires every optimum to have the same wave or the wave boundaries to coincide with the endpoints of C .

10 Applications, implementation, and discussion

The preceding results separate two roles for staged access. Finite rationing changes continuation lotteries and can screen heterogeneous agents, whereas the shape of an atomless value distribution determines whether a deterministic rollout should move smoothly or in waves. This section connects those forces to the applications and discusses when commitment to dated capacity is plausible. The continuous rationing-flow construction linking repeated eligibility to the separating ridge appears in the Appendix.

10.1 Applications

The model is most natural when timing matters, prior use generates a public signal, and the institution can commit to an observable access schedule. In each application below, the private value is the agent's payoff from adoption in the good state, the bad-news process summarizes adverse evidence generated by prior use, and the principal minimizes the time needed to reach a prespecified coverage target below the full population.

Vaccination and public health. A health authority is the principal, eligible individuals are the agents, and type captures the private benefit of a successful vaccine relative to its perceived downside. Earlier vaccinations generate public safety and effectiveness evidence; eligibility rules and appointment capacity implement dated scarcity. The completion target is a coverage objective below universal take-up. The theorem-based prediction concerns the effect of credible access timing on strategic delay.¹⁰

Agricultural extension. A government or NGO controls demonstration slots or subsidized access to a seed, fertilizer, pest-control technology, or insurance product. Farmers differ in the private return from successful adoption and learn from earlier users. A dated quota can bring experimentation forward without introducing a price, while the target represents the penetration level the program seeks to reach within the relevant horizon.

Platforms and digital products. A platform or artificial-intelligence provider controls invitations, waitlist clearances, or capacity tiers. Users differ in the value of a successful product and update from public experience, reviews, and observed failures. The service lottery maps into a queue or access lottery, and the target is an early penetration or critical-mass objective. Here the commitment assumption is most plausible when the queue and its clearing rule are publicly observable.¹¹

10.2 Why commitment is plausible

In public programs, production capacity, procurement contracts, certified logistics, and announced priority rules constrain cumulative releases. In technology markets, deviations from a public waitlist are visible to users and the press, and can destroy the attention or trust the waitlist was intended to create. Repeated institutional rollouts add reputational discipline. These mechanisms differ across applications, but they all make the cumulative path more credible than a cheap-talk promise to ration.

Where no such commitment channel exists, the relevant object is a no-commitment timing equilibrium rather than the optimum characterized here. The gap between committed and Markov-perfect rollout is a natural extension.

10.3 Comparative statics and ethical interpretation

The model delivers several robust comparative statics at the level of feasible paths. Increasing β accelerates any fixed adoption path's belief movement and weakly relaxes every lower-type entry condition. Increasing the mass of early adopters has the same local effect. Increasing r makes delayed opportunities less attractive, which relaxes upward incentive constraints, but it

¹⁰The model does not rank the speed objective against clinical priority, equity, or epidemiological externalities. The vaccination discussion therefore interprets one incentive channel rather than offering a complete policy prescription.

¹¹The source of staged access may be physical capacity, deliberate invitation design, or both. The model takes the committed cumulative access path as the principal's instrument and asks how its timing changes users' incentives.

also changes the separating ridge and can reduce the amount of experimentation that is privately sustained. Distributional changes are more subtle because they move both the stock generating information and the marginal type that must be persuaded.

Artificial scarcity in health settings raises an obvious ethical concern. The model is not an argument for withholding units that could be administered safely and beneficially. It establishes that perceived future availability is part of the strategic environment and that quantity announcements can affect take-up. The relevant policy conclusion is comparative: when a government already faces physical scarcity, or when the target is close and future access is one source of delay, the order and credibility of eligibility waves can matter for speed. Clinical priority, equity, and external health benefits remain outside the objective studied here and should be combined with, not replaced by, the mechanism identified in the paper.

The same concern is weaker in digital and platform applications, where a waitlist reallocates access over short horizons rather than withholding a medical treatment. There the model suggests a design principle: an early lottery can create experimentation without granting a broad right to immediate service, while a later clearing wave prevents the waitlist from becoming so slow that learning itself stalls.

11 Discussion and conclusion

When adoption produces information, abundant supply need not deliver rapid diffusion. Forward-looking agents value the option to wait for others, and dated scarcity changes that option by making future access contingent rather than guaranteed.

The finite results show how this change creates a dynamic screening problem. With one value, scarcity eliminates within-class delay. With two target-relevant values, an initial and a terminal batch attain the unrestricted optimum. At three values, however, an early rationed opportunity and a later clearing opportunity can perform different incentive tasks. Rejection-contingent continuation is then itself a screening instrument, and the optimal number of releases need not equal the number of values.

This failure of a universal batch-count bound does not eliminate finite structure. For every fixed release bound, an optimum exists and can be represented with all retained releases immediately exhausted. Upper-block willingness continues to order participation, while continuation-value dominance identifies when an earlier rationed opportunity cannot separate a residual block and may be removed. The mass-gap condition provides a primitive sufficient test for that conclusion.

The continuum analysis reveals a second structural layer. Conditional on schedule-valued service, the prior and target pin down compulsory initial and terminal batch cores. Continuous separation follows an explicit ridge, while every departure from that ridge is a pooled excursion. For a general smooth distribution, boundary-attached excursions can enlarge the observed boundary batches. With a positive nonincreasing density, every nondegenerate excursion delays completion, so the unique optimum in this class is an initial batch followed by smooth rollout and a terminal batch. Sufficiently convex active regions create the opposite local force, and global compatibility determines whether that gain appears as a wave.

The finite approximation results give a more modest connection to discrete economies. Strongly regular continuum schedules can be approached by finite direct menus, and terminal padding shows that the calendar-count advantage can vanish along terminal-rich refinements even when exact finite simplicity fails. These are one-directional links and do not identify unrestricted finite optimal plans with the direct-scheduling problem.

Taken together, the results give a unified view of staged access. Scarcity can accelerate experimentation by limiting the option to postpone, rationing can screen agents through continuation after rejection, and the distribution of values determines whether schedule-valued rollout separates smoothly or advances in pooled waves.

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Appendix

This appendix contains the proofs, numerical verification, and auxiliary constructions cited in the main text. Result numbers are section-prefixed in the combined file; source labels provide clickable cross-references from the main manuscript. The deterministic-schedule decomposition and the nonincreasing-density optimum appear in Sections A.9–A.10. Throughout, “service” means successful allocation after any rationing, and rejected applicants remain eligible unless a result states otherwise. The deterministic continuum results assign one certain service date to almost every value and do not assert purification of lottery-valued relaxed limits.

A.1 Operational equilibrium and lottery values

At date t , anonymous rationing gives acceptance probability

$$Q_t = \begin{cases} 0, & M_t^- = S_t, \\ 1, & M_t^- < S_t \text{ and } A_t \leq S_t - M_t^-, \\ (S_t - M_t^-)/A_t, & A_t > S_t - M_t^- > 0. \end{cases}$$

Rejected agents remain active. A realized application plan α specifies a sequence of dates contingent on the agent’s own past failures. Conditional on the no-signal path, the probability of first success at date $s \in T^\alpha$ is

$$Q_s \prod_{k \in T^\alpha: k < s} (1 - Q_k).$$

The normalized time-zero value is therefore

$$U(v; \alpha) = \sum_{s \in T^\alpha} Q_s \prod_{k < s} (1 - Q_k) e^{-rs} (v - B_s).$$

This formula is affine in v . For any two types $v > v'$, the difference between their values from one fixed lottery is

$$(v - v') \sum_s Q_s \prod_{k < s} (1 - Q_k) e^{-rs}.$$

This is the discounted-service moment that underlies upper-block willingness and the averaged-envelope interpretation.¹²

Lemma A.1.1 (Upper-block willingness). *Fix any rejection-contingent continuation. If type v' weakly prefers applying now to that continuation, every type $v > v'$ strictly prefers applying now, unless immediate acceptance probability is zero.*

Proof. Let A^+ be the continuation lottery after rejection and write its payoff as $av - c$. Applying now with probability $Q > 0$ gives

$$Qe^{-rt}(v - B_t) + (1 - Q)(av - c).$$

¹²The affine-in-type comparison is the familiar single-crossing logic of screening models; see, for example, Myerson (1981).

The difference from rejecting is $Q[(e^{-rt} - a)v - (e^{-rt}B_t - c)]$. If the difference is nonnegative at v' , it is strictly larger at $v > v'$ because every possible service date in A^+ is later than t , so $a < e^{-rt}$. $\square$

A.2 Free supply

Under free supply, successful service occurs at the first application. Let $g_v(t) = e^{-rt}(v - B_t)$. Differentiation gives

$$\dot{g}_v(t) = e^{-rt}\{\beta M_t B_t - r(v - B_t)\}.$$

On an interior mixed-stopping interval, $\dot{g}_v = 0$, hence

$$B_t = \frac{rv}{r + \beta M_t}.$$

Differentiating the right-hand side and using $\dot{B}_t = -\beta M_t B_t$ yields

$$\dot{M}_t = M_t(r + \beta M_t).$$

The unique solution beginning at $M_a > 0$ is

$$M_t = \frac{rM_a e^{r(t-a)}}{r + \beta M_a(1 - e^{r(t-a)})},$$

up to the time at which the class is exhausted. Its second derivative is positive.

For a continuous distribution, let $v_t = F^{-1}(1 - M_t)$. The marginal condition is

$$v_t = B_t \left(1 + \frac{\beta M_t}{r}\right).$$

For a uniform distribution, substitution of $v_t = \bar{v} - (\bar{v} - \underline{v})M_t$ gives an autonomous scalar equation. The curvature of its solution changes sign at most once. The explicit threshold conditions in the main text follow by evaluating the derivative at the two support boundaries.

A.3 One-value unraveling

Release exactly $\bar{M} < 1$ at zero. Suppose a selected equilibrium reaches the target at $T > 0$. There must be a first successful adoption date $t_0 \leq T$. If $t_0 > 0$, beliefs are unchanged on $[0, t_0)$ and an agent scheduled at t_0 strictly prefers applying at zero. Thus $t_0 = 0$.

If capacity remains before T , any agent scheduled at a later date can apply slightly earlier and obtain certain service with less discounting and no worse belief. If capacity is exhausted before T , the target has already been reached because total supply equals $\bar{M}$. If instead the target is completed by a positive rationed mass at T , an applicant can move slightly earlier and face weakly more available stock before the terminal rush. Every delayed outcome is therefore contradicted. Immediate completion is optimal.

A.4 Two-value optimum

Let high mass be q_H . Any feasible plan completing at T must satisfy the exposure bound

$$\beta \int_0^T M_t dt \leq \beta q_H T$$

before low-type entry, because no low type has yet adopted and at most all high types can contribute. Since the earliest low-type participation has $B_T = v_L$,

$$\log \frac{B_0}{v_L} = \beta \int_0^T M_t dt \leq \beta q_H T.$$

This gives $T \geq T_E$.

A high type assigned positive probability of service at zero must obtain at least the value of waiting for the terminal certain opportunity. The fastest terminal belief is v_L , so

$$v_H - B_0 \geq e^{-rT}(v_H - v_L),$$

which gives $T \geq T_H$.

Set $T^* = \max\{T_E, T_H\}$ and

$$m_0^* = \frac{1}{\beta T^*} \log \frac{B_0}{v_L}.$$

Release m_0^* at zero. Keep the stock fixed until T^* , when belief equals v_L , and release the remaining target mass. The low type is willing. The high type's terminal deviation is weakly unprofitable by construction. If $m_0^* < q_H$, implement the initial mass by rationing all high agents or by deterministic splitting. This attains both lower bounds and proves optimality.

A.5 Three-value construction

The primitives are

$$(v_1, v_2, v_3) = (0.86, 0.8, 1/3), \quad (q_1, q_2, q_3) = (0.2, 0.4, 0.4),$$

$$B_0 = 0.75, \quad \beta = 0.5, \quad r = 0.4, \quad \bar{M} = 0.98.$$

The policy parameters are

$$s = 1.7489035062, \quad t = 2.2062097487, \quad Q = 0.1104189201, \quad T = 4.1402437122.$$

Beliefs are

$$B_s = 0.75e^{-0.1s} = 0.6296618038,$$

$$B_t = B_s e^{-0.5(0.2+0.4Q)(t-s)} = 0.5954712920,$$

$$B_T = B_t e^{-0.3(T-t)} = 1/3.$$

The service payoffs are

$$\begin{aligned} g_1(0) &= 0.11, \\ g_1(s) &= 0.1144327420, \\ g_1(t) &= 0.1094497875, \\ 0.95g_1(T) &= 0.0955048525, \end{aligned}$$

so

$$Qg_1(s) + (1 - Q)g_1(t) = 0.11.$$

For type 2,

$$g_2(s) = g_2(t) = 0.95g_2(T) = 0.0846245529.$$

Type 3 obtains negative service payoff at s and t and zero at T . These calculations establish sequential feasibility under favorable tie selection.

A.5.1 A finite binding-case analysis for three-type simple plans

A simple plan has at most three immediately exhausted releases. Normalize the first positive release to date zero and denote the other possible dates by $s < T$. If type 3 applies at a preterminal date, upper-block willingness implies that every residual higher type also applies. Increasing that release until the target is reached only raises the acceptance probability of current applicants and completes strictly earlier. Hence type 3 appears only at T in an optimal simple plan.

At each of the two preterminal dates, the applicant set is therefore an upper block among types 1 and 2, with at most one partial cutoff. There are four possible positive-demand modes:

$$(1, P), \quad (1, F), \quad (2, P), \quad (2, F),$$

where $(1, P)$ means that a positive proper fraction of residual type 1 applies, $(1, F)$ that all residual type 1 applies and type 2 does not, $(2, P)$ that all residual type 1 and a positive proper fraction of residual type 2 apply, and $(2, F)$ that all residual types 1 and 2 apply. Zero service is included as a boundary. Thus there are at most $4^2 = 16$ ordered preterminal mode pairs.

For a fixed pair, let $a_{nk} \in [0, 1]$ be the application share of residual type n at date $k \in \{0, s\}$, let Q_k be the common anonymous acceptance probability, and let R_{nk} be the residual mass immediately before the date. Service is

$$y_{nk} = Q_k a_{nk} R_{nk}, \quad R_{n,k+} = R_{nk} - y_{nk}.$$

Beliefs and the terminal acceptance probability are determined by the realized service masses. Write

$$u_n(0) = v_n - B_0, \quad u_n(s) = e^{-rs}(v_n - B_s), \quad W_n^T = Q_T e^{-rT}(v_n - B_T).$$

At the middle date the mode conditions are

$$\begin{aligned} a_{ns} = 0 &\implies u_n(s) \leq W_n^T, \\ 0 < a_{ns} < 1 &\implies u_n(s) = W_n^T, \\ a_{ns} = 1 &\implies u_n(s) \geq W_n^T. \end{aligned}$$

The continuation value before the middle date is

$$V_n^s = W_n^T + Q_s a_{ns} \{u_n(s) - W_n^T\}.$$

At date zero the corresponding conditions are

$$\begin{aligned} a_{n0} = 0 &\implies u_n(0) \leq V_n^s, \\ 0 < a_{n0} < 1 &\implies u_n(0) = V_n^s, \\ a_{n0} = 1 &\implies u_n(0) \geq V_n^s. \end{aligned}$$

For each mode pair, the equalities associated with partial cutoffs, together with terminal participation, determine a finite-dimensional candidate; the remaining inequalities accept or reject it. Comparing the accepted candidates and their boundaries gives the global simple-plan optimum. This is the corrected version of the 2024 three-type algorithm: the old verbal cases are replaced by a complete enumeration of binding cases that retains rejection-contingent lotteries.

A.5.2 Eliminating noncompetitive modes in the counterexample

For the numerical economy, every simple plan that can challenge the four-release construction lies in one reduced family.

First, suppose type 2 applies at date zero. Its continuation value from not applying must then be no larger than

$$v_2 - B_0 = 0.05.$$

At completion, type 3 participation implies $B_T \leq v_3$. Preterminal adoption is at most $q_1 + q_2 = 0.6$, so

$$Q_T = \frac{\bar{M} - m}{1 - m} \geq 0.95.$$

The terminal option available to type 2 therefore yields

$$0.05 \geq 0.95e^{-0.4T}(v_2 - v_3),$$

and hence

$$T \geq \frac{1}{0.4} \log \frac{0.95(v_2 - v_3)}{0.05} = 5.4557473178. \tag{A.1}$$

Such a mode cannot be optimal.

Second, suppose no type-2 mass is served before T . Then $M_t \leq q_1 = 0.2$ for every $t < T$. Since type 3 must be willing at completion,

$$\log \frac{B_0}{v_3} \leq \beta \int_0^T M_t dt \leq \beta q_1 T,$$

so

$$T \geq \frac{\log(0.75/(1/3))}{0.5(0.2)} = 8.1093021622. \quad (\text{A.2})$$

Thus every competitive simple plan serves only type 1 at zero and has type 2 enter at the middle date.

Let $x \in [0, 0.2]$ be type-1 service at zero. At s , every residual type 1 applies whenever type 2 does. We may also take all type-2 agents to apply. To see this, begin from a mode in which only a share $\alpha < 1$ of type 2 applies. Partial application requires type-2 indifference between service at s and terminal continuation. Increase the applicant share to one and reduce the common acceptance probability so that total middle service is unchanged. The belief path, terminal date, and terminal acceptance probability are unchanged. Type 2 remains indifferent, while type 1 receives a lower probability of the relatively attractive middle service and therefore has a weaker deviation. Hence full type-2 application is without loss.

Let p denote the common middle acceptance probability. Total preterminal adoption, the middle belief, the terminal date, and terminal acceptance are

$$m = x + p(0.6 - x), \quad (\text{A.5.1})$$

$$B_s = 0.75e^{-0.5xs}, \quad (\text{A.5.2})$$

$$T(x, s, p) = s + \frac{\log(B_s/v_3)}{0.5m}, \quad (\text{A.5.3})$$

$$Q_T(x, s, p) = \frac{0.98 - m}{1 - m}. \quad (\text{A.5.4})$$

Define

$$g_n = e^{-0.4s}(v_n - B_s), \quad h_n = Q_T e^{-0.4T}(v_n - v_3), \quad C_n = pg_n + (1 - p)h_n.$$

The reduced simple-plan problem is exactly

$$\min_{0 \leq x \leq 0.2, s \geq 0, 0 \leq p \leq 1} T(x, s, p) \quad (\text{SP})$$

subject to

$$g_2 \geq h_2, \quad (\text{A.5.5})$$

$$C_1 \leq 0.11, \quad (\text{A.5.6})$$

$$C_2 \geq 0.05. \quad (\text{A.5.7})$$

These three inequalities have direct meanings: type 2 applies at s , type 1 accepts date-zero service, and type 2 does not imitate the date-zero opportunity. Type 3's earlier constraints are automatic because $B_s > v_3$ whenever $T > s$ and $B_T = v_3$.

A.5.3 Solving the reduced problem

For fixed (x, s) , completion is strictly decreasing in p :

$$\frac{\partial T}{\partial p} = -\frac{(0.6 - x) \log(B_s/v_3)}{0.5m^2} < 0. \quad (\text{A.3})$$

On the sublevel set $T \leq 4.1701$, terminal continuation h_n is strictly increasing in p . Indeed,

$$\frac{\partial \log h_n}{\partial m} = \frac{0.98 - 1}{(0.98 - m)(1 - m)} + \frac{0.4 \log(B_s/v_3)}{0.5m^2} > 0 \quad (\text{A.4})$$

throughout the relevant compact region. In addition, $g_1 > h_1$ whenever $g_2 \geq h_2$, because

$$g_1 - h_1 = (g_2 - h_2) + (v_1 - v_2) (e^{-0.4s} - Q_T e^{-0.4T}) > 0.$$

Hence C_1 rises with p , while the type-2 participation slack $g_2 - h_2$ falls. The maximal feasible p is therefore determined by one of

$$g_2 = h_2, \quad C_1 = 0.11, \quad p = 1.$$

Condition (A.5.7) is slack on every plan with $T \leq 4.1701$, since $C_2 \geq h_2 \geq 0.95e^{-0.4(4.1701)}(v_2 - v_3) > 0.0836$.

Substitution of each binding case leaves a smooth one-dimensional branch. Elementary implicit differentiation gives the following exhaustive comparison:

Binding case	Minimizing behavior	Best completion
$g_2 = h_2, C_1 < 0.11$	T decreases as x rises until C_1 binds	> 4.1700003382
$g_2 = h_2, C_1 = 0.11$	The lower branch decreases in x ; the upper branch is slower	4.1700003382
$p = 1$	Type-2 participation binds at the optimum	4.2412342351
Remaining boundaries	Covered by (A.1), (A.2), or a preceding row	> 4.1700003382

The derivative numerators have no zero on their indicated compact branch intervals; clearing their positive denominators gives the signs in the table. In particular, every minimizing branch is improved by raising x until

$$x = q_1 = 0.2. \quad (\text{A.5})$$

Thus all type 1 is served at zero.

Set $x = 0.2$ and let

$$A := \frac{v_1 - v_3}{v_2 - v_3} = \frac{79}{70}, \quad B_s = 0.75e^{-0.1s}.$$

At the interior optimum, (A.5.5) and (A.5.6) bind. Using type-2 indifference to eliminate the terminal payoff from type 1's constraint gives

$$p(s) = \frac{(v_1 - B_0)e^{0.4s} - A(v_2 - B_s)}{(v_1 - B_s) - A(v_2 - B_s)}. \quad (\text{A.5.8})$$

The denominator is positive because it equals

$$\frac{(v_1 - v_2)(B_s - v_3)}{v_2 - v_3}.$$

Type-2 indifference also implies

$$T - s = \frac{1}{0.4} \log \frac{Q_T(p)(v_2 - v_3)}{v_2 - B_s}.$$

Equating this expression to the learning duration yields the scalar equation

$$\frac{\log(B_s/v_3)}{0.5[0.2 + 0.4p(s)]} = \frac{1}{0.4} \log \frac{Q_T(p(s))(v_2 - v_3)}{v_2 - B_s}. \quad (\text{A.5.9})$$

On the feasible interval, (A.5.9) has exactly two roots, isolated by monotonicity on the intervals separating its stationary points:

$$\begin{aligned} s_- &= 1.0624057254, & p_- &= 0.6044630754, & T_- &= 4.2525967757, \\ s_+ &= 1.8812680292, & p_+ &= 0.8605859690, & T_+ &= 4.1700003382. \end{aligned}$$

The second root is the minimizer. On the boundary $p = 1$, type-2 participation binds at

$$s = 2.3072002715, \quad T = 4.2412342351.$$

Therefore

$$T_{\text{simple}}^* = 4.1700003382.$$

Theorem A.5.1 (Global simple-plan benchmark for the counterexample). *For the primitives of the main-text counterexample, the minimum completion time over all plans with at most three immediately exhausted releases is*

$$4.1700003382.$$

It is attained by serving all type 1 at zero, accepting the fraction 0.8605859690 of type 2 at date 1.8812680292, and pooling rejected type 2 with type 3 at date 4.1700003382. Since the feasible four-release plan completes at 4.1402437122, no optimal plan is simple.

Proof. Upper-block willingness and the no-preterminal-type-3 argument reduce the simple class to the finite binding-case analysis. Bounds (A.1) and (A.2) eliminate every mode with type 2 at zero or without preterminal type-2 service. The full-application transformation reduces all remaining modes to problem (SP). Equations (A.3)–(A.5) and the comparison across binding cases place the minimum at $x = 0.2$ and reduce it to the roots of (A.5.9) and the $p = 1$ boundary.

The displayed root and boundary comparisons give the claimed minimum. The strict comparison with the feasible four-release completion time proves the final statement. $\square$

A.6 Fixed- K existence and exhaustion

Fix a horizon upper bound $\bar{T}$ obtained from any feasible policy. Permit K weakly ordered dates in $[0, \bar{T}]$ and zero-capacity dummy releases. Let y_{nk} be realized service mass of type n at date k , a_{nk} applicant mass, and z_{nk} residual mass. Beliefs are continuous functions of dates and cumulative realized service. Replace acceptance ratios by the identities

$$y_{nk} = Q_k a_{nk}, \quad 0 \leq Q_k \leq 1,$$

with aggregate capacity constraints and complementarity. Sequential optimality is a finite family of weak inequalities comparing application and continuation. The feasible set is closed and bounded. Completion time is lower semicontinuous. Weierstrass gives a minimizer.

If release k contains unused inventory, reduce cumulative supply to realized service until the next release. No on-path acceptance probability or adoption changes. Repeating gives immediate exhaustion at every retained positive release. The operation postpones slack inventory; it does not consolidate distinct exhausted releases.

A.7 When should types be pooled?

A.7.1 Primitive mass-gap condition

Consider a residual block $v_k > \dots > v_j$ that can obtain certain service at a clearing or terminal date T . Suppose $B_T = v_j$ and cumulative adoption before any candidate separating opportunity is at least m . For every residual type $v_i \leq v_k$, define the time-zero payoff from certain service by

$$g_i(t) = e^{-rt}(v_i - B_t).$$

Because $M_t \geq m$ and beliefs decline to $B_T = v_j$, for every $t < T$,

$$\begin{aligned} g'_i(t) &= e^{-rt}\{(r + \beta M_t)B_t - rv_i\} \\ &\geq e^{-rt}\{(r + \beta m)v_j - rv_k\}. \end{aligned}$$

Under $\beta m v_j \geq r(v_k - v_j)$, the last expression is nonnegative. Hence $g_i(t) \leq g_i(T)$ for every $t < T$. Rejecting an earlier opportunity preserves the option of certain service at T , so $W_i(t^+) \geq g_i(T) \geq g_i(t)$. The envelope-dominance argument below therefore applies simultaneously to the whole block. If types $1, \dots, k-1$ have already been completely served, substitute $m = \sum_{\ell < k} q_\ell$.

A.7.2 General envelope dominance

Let type j enter at T and consider a residual higher type $i < j$. At $t < T$, applying with probability Q and continuing after rejection yields

$$Qg_i(t) + (1 - Q)W_i(t^+).$$

If $g_i(t) \leq W_i(t^+)$, this value is weakly below continuation for every $Q \in [0, 1]$. Hence no pre- T rationed opportunity can strictly separate type i . At T , upper-block willingness implies that every residual higher type weakly applies whenever type j does. Pooling all residual types at T preserves their incentives and weakly increases contemporaneous adoption.

On a constant-stock interval,

$$g'_i(t) = e^{-rt} \{ \beta M B_t - r(v_i - B_t) \}.$$

Since B_t falls, the expression crosses zero at most once. The envelope maximum is therefore attained at the ridge belief $rv_i/(r + \beta M)$ when that belief lies in the interval, and otherwise at an endpoint.

A.8 Finite foundations and finite approximation

This section separates three logically distinct statements: the maintained selected-sequence foundation, a sufficient direct type-pure condition, and a conditional finite-approximation theorem for ordered direct schedules.

A.8.1 Convergence and local-expansion conventions

The analytical tools used below are elementary, but two conventions are worth stating because they preserve the economic objects of interest.

First, ordinary pointwise convergence can lose a pooled jump. We therefore represent a nondecreasing right-continuous schedule h by its graph together with the vertical segment joining $h(x^-)$ and $h(x)$ at each jump. We call this the *completed graph* and use Hausdorff distance between these compact sets. For bounded monotone schedules, Helly selection gives a subsequence converging at every continuity point; with the vertical jump segments retained, this is equivalent to convergence of completed graphs. The convention is used only to keep pooled service and terminal batches visible in the limit.¹³

Second, the finite-approximation argument solves finitely many smooth contact equations. The proof displays the relevant Jacobian. When that Jacobian is bounded away from singularity on the compact set under consideration, the ordinary implicit-function theorem gives locally unique nearby contacts; a finite cover makes the bound uniform.¹⁴

¹³Related Helly-selection arguments for monotone strategies and allocations appear in [Lauermann and Speit \(2023\)](#) and [Dilmé and Garrett \(2026\)](#). For the completed-graph convention for paths with jumps, see [Whitt \(2002\)](#).

¹⁴For a standard parameterized statement of the implicit-function theorem, see [Krantz and Parks \(2002\)](#). No infinite-dimensional result is used here.

Finally, the reflected-contact expansion needs an exact way to pair the two sides of a strict local maximum. We construct that change of variable directly in Lemma A.10.4: the loss from the maximum is written as $z^2 A(p, z)$ with $A > 0$, and $z\sqrt{A(p, z)}$ turns equal-height reflection into ordinary sign reversal.¹⁵

A.8.2 Completed-graph compactness and path convergence

Let F_N have ordered quantile cells $I_{j,N}$ of masses $q_{j,N}$ and let $\Delta_N = \max_j q_{j,N} \rightarrow 0$. Outside an exceptional set of mass $o(1)$, represent the selected ordered service assignment by a nondecreasing right-continuous function τ_N on served quantile ranks and its completed graph. Suppose completion times are bounded by $\bar{T}$.

Lemma A.8.1 (Completed-graph compactness). *A subsequence of completed graphs converges in the Hausdorff metric to the completed graph of a nondecreasing correspondence τ . Right-continuous representatives converge at every continuity point and in L^1 . Their generalized inverses M_N converge almost everywhere to M , exposure processes converge uniformly, and therefore $B_N \rightarrow B$ uniformly on $[0, \bar{T}]$.*

Proof. Helly selection gives pointwise convergence of monotone representatives at continuity points; under the completed-graph convention just stated, monotonicity makes this equivalent to Hausdorff convergence of the completed graphs. Boundedness gives L^1 convergence. Generalized inverses converge at every continuity point of M and hence almost everywhere. Dominated convergence yields

$$\int_0^{\bar{T}} |M_N(s) - M(s)| ds \rightarrow 0.$$

Consequently

$$\sup_{t \leq \bar{T}} \left| \int_0^t (M_N - M) ds \right| \leq \int_0^{\bar{T}} |M_N - M| ds \rightarrow 0.$$

The exponential belief law then gives uniform belief convergence. $\square$

Completion-time convergence is not implied by graph convergence alone. We use the following endpoint condition.

Definition A.8.1 (Regular target crossing). A limiting adoption path with completion date T crosses the target regularly if $M(t) < \bar{M}$ for every $t < T$ and either: (i) M is continuous at T and crosses the target from below at a positive one-sided rate: there exist $c, \delta > 0$ such that $M(T) - M(t) \geq c(T - t)$ for every $t \in [T - \delta, T]$; or (ii) M has a jump at T whose size is bounded away from zero along the approximating sequence. Target-aligned terminal cells are allowed when their service dates converge to T .

Under this condition, completed-graph convergence implies convergence of hitting times. Without it, a vanishing terminal cell can determine every finite completion date while disappearing from the limiting graph.

¹⁵This explicit construction is the one-dimensional quadratic normal form often associated with the Morse lemma; the proof below does not invoke the general theorem. For background, see [Matsumoto \(2002\)](#).

A.8.3 Finite approximation of direct schedules

Assume finitely many smooth components and pools, transverse endpoint contacts, adjacent-cutoff Jacobians uniformly bounded away from zero, and uniform slack in every nonbinding global direct-IC and lower-tail inequality.

Proposition A.8.1 (Finite approximation of strongly regular schedules). *Under these assumptions, target-aligned adjacent quantile partitions admit whole-cell finite direct menus with exact adjacent cutoff equalities, exhausted batches, global direct IC, and completion times converging to the continuum schedule.*

Proof. Assign each separating cell the date of its marginal continuum rank and assign all cells in a pooled block to the common pool date. The residual adjacent indifference equations and pool-contact equations form a finite system whose Jacobian converges to the block-diagonal continuum Jacobian. Uniform nonsingularity and the implicit-function theorem give date corrections tending to zero. Compactness turns the local bounds into a uniform one over the finitely many component types. Affine single crossing turns exact adjacent equalities into global ordering; uniform slack preserves every nonbinding comparison and lower-tail inequality. Aggregate graphs, exposure, beliefs, and completion times converge. $\square$

This proposition is conditional and concerns ordered direct menus. It does not prove that every such menu is sequentially implementable in the original repeated-application game.

A.9 Deterministic service schedules and exact excursion decomposition

Let $m \in [0, \bar{M}]$ index served agents from highest to lowest payoff, so the marginal payoff is $\nu(m) = F^{-1}(1 - m)$. An ordered deterministic direct schedule is a weakly increasing right-continuous map $\tau : [0, \bar{M}] \rightarrow [0, T]$ assigning rank m the point-mass service distribution $\delta_{\tau(m)}$. Flat intervals of τ pool different values at one certain date. By contrast, a pure contingent application strategy in the operational model may still induce a nondegenerate service-time lottery after rationing and rejection; such lottery-valued allocations are excluded from this section. Let λ denote Lebesgue measure on $[0, \bar{M}]$. The schedule induces

$$M_t = \lambda\{m \in [0, \bar{M}] : \tau(m) \leq t\}, \quad B_t = B_0 \exp \left\{ -\beta \int_0^t M_s ds \right\}.$$

For each service date t , write

$$L_t(v) = e^{-rt}(v - B_t).$$

Global direct IC means that $L_{\tau(m)}(\nu(m)) \geq L_{\tau(n)}(\nu(m))$ for every served m, n , together with participation for served types and exclusion for types below $\nu(\bar{M})$.

Define

$$b(m) = \frac{r\nu(m)}{r + \beta m}, \quad x(m) = -\log b(m),$$

and

$$\mathcal{S}(u, v) = \frac{1}{\beta} \int_u^v \frac{x'(m)}{m} dm.$$

Because ν is positive and strictly decreasing, b is strictly decreasing.

A.9.1 Boundary states and continuous service

Lemma A.9.1 (Boundary normalization and terminal zero rent). *Suppose $\bar{M} \in (0, 1)$ and*

$$\nu(\bar{M}) < B_0 < \nu(0).$$

Let τ be a finite-completion globally incentive-compatible ordered deterministic schedule serving exactly the upper $\bar{M}$ mass, with participation for served types and exclusion below $\nu(\bar{M})$. Put $t_0 = \tau(0)$ and let $T = \tau(\bar{M})$.

Subtracting t_0 from every service date preserves feasibility and reduces completion time by t_0 . Also,

$$B_T = \nu(\bar{M}). \tag{A.9.1}$$

Consequently there are unique ranks $0 < m_0 < m_1 < \bar{M}$ satisfying

$$b(m_0) = B_0, \quad b(m_1) = \nu(\bar{M}). \tag{A.9.2}$$

Proof. No adoption occurs before t_0 , so beliefs remain equal to B_0 on $[0, t_0)$. Shifting every service date backward by t_0 therefore leaves the belief attached to every corresponding service line unchanged and multiplies all dated payoffs by the common positive factor e^{rt_0} . Global IC, participation, and exclusion are preserved.

Let $v_* = \nu(\bar{M})$ be the marginal served type. Its assigned service line is the terminal line, because τ is ordered. Participation gives $L_T(v_*) \geq 0$. Every $v < v_*$ is excluded, so $L_T(v) \leq 0$. Letting $v \uparrow v_*$ yields $L_T(v_*) \leq 0$. Hence $L_T(v_*) = 0$, which is (A.9.1).

Finally, $b(0) = \nu(0)$ and

$$b(\bar{M}) = \frac{r\nu(\bar{M})}{r + \beta\bar{M}} < \nu(\bar{M}).$$

Strict monotonicity of b gives unique solutions to (A.9.2). Since $B_0 > \nu(\bar{M})$, the first solution lies strictly below the second. $\square$

Lemma A.9.2 (Continuous service lies on the ridge). *Under the hypotheses of Lemma A.9.1, let μ_c denote the continuous part of the Stieltjes measure $d\tau$. At every continuity point m of τ in $\text{supp } \mu_c$,*

$$B_{\tau(m)} = b(m). \tag{A.9.3}$$

The measure μ_c is absolutely continuous and is supported on $[m_0, m_1]$. Almost everywhere on its increasing part,

$$\tau'(m) = \frac{x'(m)}{\beta m}. \tag{A.9.4}$$

In particular, τ has no singular-continuous component.

Proof. Take a continuity point $m \in \text{supp } \mu_c$. For each n , choose $a_n < m < b_n$ with $a_n, b_n \rightarrow m$ and $\tau(b_n) > \tau(a_n)$. Let c_n be the intersection of the service lines at these two dates. Their slopes are ordered, and global IC gives

$$\nu(b_n) \leq c_n \leq \nu(a_n),$$

so $c_n \rightarrow \nu(m)$. Put $\Delta_n = \tau(b_n) - \tau(a_n)$. Continuity gives $\Delta_n \rightarrow 0$. During the corresponding time interval cumulative adoption lies between a_n and b_n , hence its time average is $m + o(1)$ and

$$B_{\tau(b_n)} = B_{\tau(a_n)} \exp\{-\beta(m + o(1))\Delta_n\}.$$

The intersection formula is

$$c_n = \frac{B_{\tau(a_n)} - e^{-r\Delta_n} B_{\tau(b_n)}}{1 - e^{-r\Delta_n}}.$$

Expanding at $\Delta_n = 0$ and passing to the limit gives

$$\nu(m) = \frac{r + \beta m}{r} B_{\tau(m)},$$

which is (A.9.3).

Beliefs never exceed $B_0 = b(m_0)$ and never fall below the terminal belief $b(m_1)$. Hence (A.9.3) and strict monotonicity of b imply

$$\text{supp } \mu_c \subseteq [m_0, m_1].$$

Let E be the set of continuity points in $\text{supp } \mu_c$; the complement of E is countable and therefore μ_c -null. For $a < c$ in E , the belief law and (A.9.3) give

$$x(c) - x(a) = \beta \int_{\tau(a)}^{\tau(c)} M_s ds \geq \beta a \{\tau(c) - \tau(a)\}.$$

Since $a \geq m_0 > 0$ and x' is bounded on $[m_0, m_1]$, there is a finite constant

$$C = \frac{\sup_{[m_0, m_1]} x'}{\beta m_0}$$

such that $\tau(c) - \tau(a) \leq C(c - a)$ for all $a < c$ in E .

Fix an interval $I = (u, v) \subseteq [m_0, m_1]$. If $\mu_c(I) = 0$, there is nothing to prove. Otherwise, because E has full μ_c -measure and μ_c is atomless, one can choose $a_n, c_n \in E \cap I$ with $a_n < c_n$ such that

$$\mu_c((u, a_n]) + \mu_c([c_n, v)) \rightarrow 0.$$

Then

$$\mu_c(I) = \lim_{n \rightarrow \infty} \mu_c((a_n, c_n)) \leq \liminf_{n \rightarrow \infty} \{\tau(c_n) - \tau(a_n)\} \leq C(v - u).$$

Thus $\mu_c \ll \lambda$, ruling out a singular-continuous component.

At almost every point with $\tau'(m) > 0$, differentiating (A.9.3) and using $d[-\log B_t]/dt = \beta M_t$ gives

$$x'(m) = \beta m \tau'(m),$$

which is (A.9.4). □

Flat intervals of τ are batches. Jumps of τ are intervals of calendar time during which no new rank is served and the existing adoption stock continues to generate information. The next results identify every such jump with an actual pooled excursion.

A.9.2 Exact crossing, contact, and pooled blocks

Fix

$$e = (m_d, p, m_r), \quad m_d < p < m_r,$$

and put $a = \nu(p)$ and $k = r/(\beta p)$. The excursion leaves the ridge at m_d , immediately releases mass $p - m_d$, holds cumulative adoption fixed at p , and rejoins the ridge at m_r . Its waiting time is

$$\Delta(e) = \frac{1}{\beta p} \log \frac{b(m_d)}{b(m_r)}. \quad (\text{A.9.5})$$

The pivot type a is indifferent between the departure and reentry slots when

$$a - b(m_d) = e^{-r\Delta(e)} \{a - b(m_r)\}. \quad (\text{A.9.6})$$

Equivalently, with

$$q_p(m) = \log \left[b(m)^k \{a - b(m)\} \right],$$

the contact condition is

$$q_p(m_d) = q_p(m_r). \quad (\text{A.9.7})$$

Lemma A.9.3 (Exact excursion feasibility). *Suppose ν is positive, continuously differentiable, and strictly decreasing on a neighborhood of $[m_d, m_r]$. If (A.9.7) holds and the contact state lies in the interior ridge range, then the excursion clock is uniquely given by (A.9.5). The departure and reentry payoff lines cross exactly at $a = \nu(p)$. If the adjacent intersections before departure and after reentry are ordered, the complete dated menu is globally incentive compatible for all participating types.*

Proof. Strict monotonicity of b makes (A.9.5) the unique positive clock connecting the two ridge beliefs while cumulative adoption is fixed at p . Substituting this clock into the equality of the two dated payoffs gives (A.9.6), so their unique affine intersection is a .

Every dated slot generates an affine line $L_j(v) = e^{-rt_j}(v - B_j)$ whose slope decreases strictly with its date. If c_j is the intersection of L_j and L_{j+1} and $c_1 > \dots > c_{J-1}$, then

$$L_j(v) - L_{j+1}(v) = (e^{-rt_j} - e^{-rt_{j+1}})(v - c_j)$$

shows that the lines form the upper envelope in the required order. Apply this observation to the smooth prefix, departure line, reentry line, and smooth continuation. □

Lemma A.9.4 (Every waiting interval has virtual ridge endpoints). *Under the hypotheses of Lemma A.9.1, let $p \in (0, \bar{M})$ be a jump point of τ , and write*

$$t^- = \tau(p^-), \quad t^+ = \tau(p), \quad \Delta = t^+ - t^- > 0.$$

Let $B^- = B_{t^-}$ and $B^+ = B_{t^+}$. There are unique $d < p < h$ such that

$$B^- = b(d), \quad B^+ = b(h),$$

and the clock and crossing equations of the exact excursion (d, p, h) hold.

Proof. During the wait cumulative adoption is p , so

$$B^+ = B^- e^{-\beta p \Delta}. \tag{A.9.8}$$

Taking one-sided limits of the IC inequalities at rank p makes the adjacent service lines intersect at $a = \nu(p)$:

$$a - B^- = e^{-r\Delta} \{a - B^+\}. \tag{A.9.9}$$

The right-continuous convention assigns rank p to the later line. Its participation constraint and (A.9.9) imply $a - B^+ > 0$ and $a - B^- > 0$.

By Lemma A.9.1, every service belief lies in $[b(m_1), b(m_0)]$. Strict monotonicity of b therefore gives unique $d, h \in [m_0, m_1]$ with the stated beliefs. Put $k = r/(\beta p)$. Equations (A.9.8)–(A.9.9) imply

$$(B^-)^k (a - B^-) = (B^+)^k (a - B^+).$$

On $(0, a)$, the function $z \mapsto z^k(a - z)$ has a unique maximum at

$$\frac{k}{k+1}a = b(p).$$

Since $B^- > B^+$ and the two function values agree, the beliefs lie on opposite sides of the maximum:

$$B^- > b(p) > B^+.$$

As b is strictly decreasing, $d < p < h$. Equations (A.9.8)–(A.9.9) are then exactly (A.9.5)–(A.9.6). $\square$

Lemma A.9.5 (A jump is implemented by its pooled blocks). *For each jump at p with virtual endpoints $d < p < h$,*

$$\tau(m) = t^- \quad \text{for } d < m < p, \quad \tau(m) = t^+ \quad \text{for } p < m < h. \tag{A.9.10}$$

If jump i precedes jump j , their virtual supports are ordered:

$$h_i \leq d_j. \tag{A.9.11}$$

Thus every jump is an actual departure batch, wait, and reentry batch, and distinct excursion supports have disjoint interiors.

Proof. If jump i precedes jump j , beliefs are weakly lower at the later jump, so

$$B_i^+ \geq B_j^-.$$

Since $B_i^+ = b(h_i)$, $B_j^- = b(d_j)$, and b is strictly decreasing, this is (A.9.11).

Consider one jump (d, p, h) . Another jump with pivot $q \in (d, p)$ would occur earlier and hence satisfy $h_q \leq d$. But every jump has $q < h_q$, contradicting $q > d$. Likewise, a jump with pivot $q \in (p, h)$ would occur later and satisfy $h \leq d_q < q$, contradicting $q < h$. Thus there are no other jumps inside either interval in (A.9.10).

There is also no continuous increase in (d, p) . If a continuity point m in that interval belonged to $\text{supp } \mu_c$, then $\tau(m) \leq t^-$ and therefore $B_{\tau(m)} \geq B^- = b(d)$. Lemma A.9.2 would instead require $B_{\tau(m)} = b(m) < b(d)$, a contradiction. The argument on (p, h) is symmetric: there $\tau(m) \geq t^+$ and hence $B_{\tau(m)} \leq B^+ = b(h)$, whereas ridge service would require $b(m) > b(h)$.

The Stieltjes measure $d\tau$ has neither atoms nor continuous mass on (d, p) and (p, h) . Hence τ is constant on each interval, and the one-sided endpoint values give (A.9.10). $\square$

A.9.3 Global decomposition of arbitrary monotone schedules

Theorem A.9.1 (Deterministic schedule decomposition). *Under the hypotheses of Lemma A.9.1, let $\bar{\tau} = \tau - t_0$ be the time-normalized schedule. Then:*

(i) $\bar{\tau}(m) = 0$ on $[0, m_0]$ and $\bar{\tau}(m) = T - t_0$ on $[m_1, \bar{M}]$;

(ii) there is a finite or countable chronological family of exact excursions

$$\mathcal{A} = \{(d_j, p_j, h_j)\}_{j \in J}$$

with pairwise disjoint support interiors in (m_0, m_1) ;

(iii) outside those supports, every increase of $\bar{\tau}$ is absolutely continuous ridge passage;

(iv) completion time satisfies

$$T = t_0 + \mathcal{S}(m_0, m_1) + \sum_{j \in J} \Gamma_F(d_j, p_j, h_j), \quad (\text{A.9.12})$$

where

$$\Gamma_F(d, p, h) = \frac{1}{\beta} \int_d^h x'(m) \left(\frac{1}{p} - \frac{1}{m} \right) dm. \quad (\text{A.9.13})$$

The series in (A.9.12) is absolutely convergent.

Proof. The normalization is feasible by Lemma A.9.1. We first establish the boundary batches. Continuous increase below m_0 is impossible: Lemma A.9.2 would require a service belief $b(m) > b(m_0) = B_0$. A jump at $p < m_0$ is also impossible. Its departure belief satisfies $B^- \leq B_0 = b(m_0)$,

so its virtual departure rank obeys $d \geq m_0$, contradicting $d < p$. Thus $d\bar{\tau}$ vanishes on $(0, m_0)$ and $\bar{\tau} = 0$ there.

Similarly, continuous increase above m_1 would require $B = b(m) < b(m_1) = B_T$, although beliefs cannot fall below their terminal value before completion. A jump at $p > m_1$ would have $B^+ \geq B_T = b(m_1)$ and therefore $h \leq m_1$, contradicting $p < h$. Hence $d\bar{\tau}$ vanishes on $(m_1, \bar{M})$ and the schedule is constant at its completion date there. This proves (i).

Lemma A.9.2 makes the continuous part of $d\bar{\tau}$ absolutely continuous and pins every positive-density increase to the ridge. The atomic part contains at most countably many jumps. Lemmas A.9.4 and A.9.5 associate each jump with an actual exact excursion, order the supports chronologically, and make their interiors disjoint. Any same-date mass at the initial line beyond m_0 is the departure block of the first excursion; any same-date mass at the terminal line below m_1 is the reentry block of the last excursion. These are merely splits of one service line and do not change dates, beliefs, payoffs, or allocations.

There is no uncovered interior batch. Indeed, let $(a, c) \subset (m_0, m_1)$ be a maximal interval on which $\bar{\tau}$ is constant and whose interior is disjoint from all excursion supports. Each endpoint is approached either by continuous ridge service or by an excursion endpoint. In both cases the belief at the endpoint is the corresponding ridge belief. Since no time passes across (a, c) , these beliefs are equal, so $b(a) = b(c)$. Strict monotonicity of b forces $a = c$. Thus every positive-mass interior flat block belongs to an excursion, and every remaining part of the interior is ridge passage. This proves (ii) and (iii).

On the complement of the excursion supports, ridge passage takes

$$\frac{1}{\beta} \int \frac{x'(m)}{m} dm.$$

Excursion (d_j, p_j, h_j) takes

$$\Delta_j = \frac{1}{\beta p_j} \{x(h_j) - x(d_j)\} = \frac{1}{\beta} \int_{d_j}^{h_j} \frac{x'(m)}{p_j} dm,$$

whereas smooth passage through the same ridge states takes $\mathcal{S}(d_j, h_j)$. Countable additivity over the disjoint supports gives (A.9.12). Finally,

$$\sum_j \Delta_j \leq T - t_0, \quad \sum_j \mathcal{S}(d_j, h_j) \leq \mathcal{S}(m_0, m_1),$$

so

$$\sum_j |\Gamma_F(d_j, p_j, h_j)| \leq T - t_0 + \mathcal{S}(m_0, m_1) < \infty.$$

□

Remark 2 (Compulsory boundary cores versus observed boundary batches). The ranks m_0 and m_1 are unique without any restriction on the sign of f' . Every admissible deterministic schedule assigns the upper block $[0, m_0]$ to its first service line and the lower target block $[m_1, \bar{M}]$ to its terminal line. These are compulsory boundary cores. They need not equal the full observed

boundary jumps: an excursion with departure rank $d = m_0$ adds its departure block to the initial batch, while an excursion with reentry rank $h = m_1$ adds its reentry block to the terminal batch. When all nondegenerate excursions are excluded, as under a nonincreasing density, the compulsory cores are exactly the full initial and terminal batches.

A.10 Diffuse and concentrated density regions

A.10.1 Nonincreasing density and the deterministic optimum

For one feasible excursion abbreviate

$$q(m) = \log\{b(m)^k(a - b(m))\}, \quad L(m) = -\frac{b'(m)}{b(m)}.$$

The contact equation gives $q(m_d) = q(m_r)$. Direct differentiation yields

$$q'(m) = L(m)R(m), \quad R(m) = \frac{(k+1)b(m) - ka}{a - b(m)},$$

so q' is positive on (m_d, p) and negative on (p, m_r) . Put

$$h(m) = \frac{1}{p} - \frac{1}{m}, \quad \psi(m) = \frac{h(m)}{R(m)},$$

where $\psi(p)$ denotes the continuous limit.

Lemma A.10.1 (Convex quantiles make ψ decreasing). *If ν is convex on the excursion interval, then ψ is strictly decreasing.*

Proof. For $m \neq p$, set $s(m) = -(\nu(m) - a)/(m - p) > 0$. Convexity makes s nonincreasing. Algebra gives $\psi = -\phi$, where

$$\phi(m, s) = \frac{\beta}{r} \frac{a\beta + r(1 - p/m)s}{a\beta + (r + \beta p)s}.$$

Writing $D = a\beta + (r + \beta p)s$,

$$\partial_m \phi = \frac{\beta p s}{m^2 D} > 0, \quad \partial_s \phi = \frac{\beta}{r} \frac{a\beta\{-rp/m - \beta p\}}{D^2} < 0.$$

Since $s'(m) \leq 0$ almost everywhere and s is nonincreasing, ϕ is strictly increasing and ψ strictly decreasing. The value at p follows by the continuous extension. $\square$

Lemma A.10.2 (Feasibility of the batch–smooth–batch schedule). *Let m_0, m_1 solve (A.9.2) and define*

$$\tau^*(m) = \begin{cases} 0, & 0 \leq m \leq m_0, \\ \int_{m_0}^m \frac{x'(z)}{\beta z} dz, & m_0 < m < m_1, \\ \mathcal{S}(m_0, m_1), & m_1 \leq m \leq \bar{M}. \end{cases} \quad (\text{A.10.1})$$

Then τ^* induces the ridge belief $B_{\tau^*(m)} = b(m)$ on $[m_0, m_1]$ and is globally incentive compatible. Served types participate, types below $\nu(\bar{M})$ prefer exclusion, and completion time is $\mathcal{S}(m_0, m_1)$.

Proof. The initial batch leaves belief unchanged at $B_0 = b(m_0)$. Along the increasing part,

$$\beta \int_{m_0}^m z d\tau^*(z) = \int_{m_0}^m x'(z) dz = x(m) - x(m_0),$$

so the induced belief is $b(m)$. At completion it is $b(m_1) = \nu(\bar{M})$; the terminal batch does not change belief instantaneously.

For $m \in [m_0, m_1]$, let

$$L_m(v) = e^{-r\tau^*(m)} \{v - b(m)\},$$

where $m = m_0$ denotes the date-zero line and $m = m_1$ the terminal line. Since

$$\tau^{*'}(m) = -\frac{b'(m)}{\beta m b(m)},$$

direct differentiation gives

$$\frac{\partial L_m(v)}{\partial m} = e^{-r\tau^*(m)} b'(m) \left[\frac{r\{v - b(m)\}}{\beta m b(m)} - 1 \right] = e^{-r\tau^*(m)} b'(m) \frac{r\{v - \nu(m)\}}{\beta m b(m)}. \quad (\text{A.10.2})$$

Because $b' < 0$ and ν is strictly decreasing, a type $v = \nu(q)$ with $q \in [m_0, m_1]$ sees $L_m(v)$ increase up to $m = q$ and decrease afterward. If $q < m_0$, it decreases throughout $[m_0, m_1]$, so the date-zero line is optimal. If $q > m_1$, it increases throughout, so the terminal line is optimal. This proves global IC.

Participation holds because $\nu(m) > b(m)$ for $m > 0$, while every rank in the terminal batch has value at least $\nu(\bar{M}) = b(m_1)$. An excluded type has $v < \nu(\bar{M})$, the terminal line is its best offered line by (A.10.2), and that line gives strictly negative payoff. Thus exclusion is optimal. The completion time follows from (A.10.1). $\square$

Theorem A.10.1 (Unique deterministic optimum under a nonincreasing density). *If f is positive and nonincreasing on the relevant valuation range, every nondegenerate exact excursion has $\Gamma_F > 0$. Under the conditions of Theorem A.9.1, the unique optimal aggregate deterministic schedule is the initial batch to m_0 , smooth ridge passage from m_0 to m_1 , and the terminal batch to $\bar{M}$.*

Proof. Because

$$\nu''(m) = -\frac{f'(\nu(m))}{f(\nu(m))^3},$$

$f' \leq 0$ makes ν convex. In addition,

$$\beta \Gamma_F = \int L(m) h(m) dm = \int q'(m) \psi(m) dm.$$

Since $q(m_d) = q(m_r)$, $\int q' = 0$. Subtracting the continuous value $\psi(p)$ gives

$$\beta\Gamma_F = \int q'(m)\{\psi(m) - \psi(p)\}dm.$$

On the left of p , both factors are positive; on the right, both are negative. Their product is positive on sets of positive measure for every nondegenerate excursion. Hence $\Gamma_F > 0$.

Theorem A.9.1 gives

$$T = t_0 + \mathcal{S}(m_0, m_1) + \sum_j \Gamma_F(e_j).$$

Therefore $T \geq \mathcal{S}(m_0, m_1)$, with equality only if $t_0 = 0$ and the excursion family is empty. Lemma A.10.2 shows that the schedule in (A.10.1) is feasible and attains this bound.

It remains to characterize equality. With no jumps and no singular-continuous component, the schedule is continuous and absolutely continuous. A nondegenerate interior flat interval is impossible: approaching its two endpoints through points carrying continuous increase, (A.9.3) would assign the same service belief to two distinct ranks, contradicting strict monotonicity of b . Hence the support of continuous increase is dense in $[m_0, m_1]$, and continuity extends (A.9.3) to the whole interval. Differentiation then gives (A.9.4) almost everywhere, which uniquely determines (A.10.1). The only remaining flat portions are the boundary batches established in Theorem A.9.1. Thus the aggregate completed graph is unique, apart from zero-mass cutoff conventions and relabeling of one same-date service line. $\square$

Remark 3 (Scope of deterministic uniqueness). Let $V^D(F)$ be the infimum of completion times over the ordered deterministic direct schedules studied in Sections A.9–A.10, and let $V^R(F)$ be the relaxed operational value allowing a disintegration K_v over service dates for almost every value. Theorem A.10.1 identifies $V^D(F) = \mathcal{S}(m_0, m_1)$ and its unique aggregate optimizer when $f' \leq 0$. It does not identify $V^R(F)$. Because the Appendix does not prove operational implementation of every abstract direct schedule, it makes no unconditional nesting claim between the two optimization domains. If the canonical deterministic schedule is operationally approximable, then $V^R(F)$ is no larger than its completion time; whether the inequality is strict remains open. Compactness of finite operational outcomes, vanishing atomic masses, and terminal padding do not by themselves imply $K_v = \delta_{\tau(v)}$ almost everywhere.

A.10.2 Local cubic coefficient and a uniform local pooling condition

Fix an interior pivot p and put $a = \nu(p)$ and $k = r/(\beta p)$. For a reflected local excursion, write

$$m_d = p - h, \quad m_r = p + \ell(h).$$

The exact contact equation is

$$q_p(p - h) = q_p(p + \ell(h)), \quad q_p(m) = \log \left[b(m)^k \{a - b(m)\} \right].$$

Lemma A.10.3 (Local reflected contact). *If ν is C^4 , positive, and strictly decreasing near p , then q_p has a strict local maximum at p . For all sufficiently small $h > 0$, there is a unique $\ell(h) > 0$ satisfying the contact equation, and*

$$\ell(h) = h - \frac{q_p'''(p)}{3q_p''(p)}h^2 + O(h^3). \quad (\text{A.10.3})$$

Proof. The function $z \mapsto z^k(a - z)$ has a strict maximum at

$$z = \frac{k}{k+1}a = b(p).$$

Since $b'(p) \neq 0$, composition with $b(m)$ gives $q_p'(p) = 0$ and $q_p''(p) < 0$. The two monotone branches around this strict maximum give a unique reflected point. Taylor expansion gives

$$q_p(p - h) = q_p(p) + \frac{1}{2}q_p''(p)h^2 - \frac{1}{6}q_p'''(p)h^3 + O(h^4),$$

$$q_p(p + \ell) = q_p(p) + \frac{1}{2}q_p''(p)\ell^2 + \frac{1}{6}q_p'''(p)\ell^3 + O(\ell^4).$$

Substitute $\ell = h + Ah^2 + O(h^3)$ and match cubic terms to obtain $A = -q_p'''(p)/(3q_p''(p))$. $\square$

Theorem A.10.2 (Cubic excursion expansion). *Let*

$$y(m) = -\frac{\nu'(m)}{\nu(m)}.$$

For the reflected excursion of Lemma A.10.3,

$$\Gamma_F(p - h, p, p + \ell(h)) = K_F(p; r, \beta)h^3 + O(h^4), \quad (\text{A.10.4})$$

where

$$K_F(p; r, \beta) = \frac{(\beta p + 2r)y(p)^2 - \beta p y'(p) + \frac{2\beta r}{\beta p + r}y(p)}{3\beta^2 p^3}. \quad (\text{A.10.5})$$

Writing $v = \nu(p)$, $f = f(v)$, and $f' = f'(v)$,

$$K_F(p; r, \beta) = \frac{2rf - \beta p v f' + \frac{2\beta r}{\beta p + r}v f^2}{3\beta^2 p^3 v^2 f^3}. \quad (\text{A.10.6})$$

Proof. Write $b_j = b^{(j)}(p)$ and $b_0 = b(p) = ra/(r + \beta p)$. Expand the exact integral (A.9.13) over $[p - h, p + \ell(h)]$, substitute (A.10.3), and collect cubic terms. The constant, linear, and quadratic differences cancel by same-state contact and reflection. Direct simplification gives

$$K_F = \frac{(\beta p + r)\{2a\beta b_1 + a\beta p b_2 + 2(\beta p + r)b_1^2\}}{3a^2\beta^2 p^3 r}. \quad (\text{A.10.7})$$

Now substitute

$$b(m) = \frac{r\nu(m)}{r + \beta m}, \quad \frac{\nu'}{\nu} = -y, \quad \frac{\nu''}{\nu} = y^2 - y'.$$

Equation (A.10.7) reduces to (A.10.5). Since

$$\nu'(m) = -\frac{1}{f(\nu(m))}, \quad y(m) = \frac{1}{\nu(m)f(\nu(m))},$$

we have

$$y'(m) = \frac{f(v) + vf'(v)}{v^2 f(v)^3}.$$

Substitution into (A.10.5) gives (A.10.6). $\square$

Lemma A.10.4 (Uniform fourth-order remainder). *Let C be a compact interior rank interval. Suppose ν is C^4 on a neighborhood of C , the reflected contact remains on the simple local branch, and all rescaled controls remain in a common compact interior feasible set. Then there are $M < \infty$ and $\bar{h} > 0$ such that*

$$|\Gamma_F(p - h, p, p + \ell_p(h)) - K_F(p)h^3| \leq Mh^4$$

for every $p \in C$ and $0 < h \leq \bar{h}$.

Proof. The ordinary inverse branches of q_p are singular at their common maximum. To remove that singularity, write the loss from the maximum in a coordinate in which equal heights are exact mirror images. Write, for z near zero,

$$q_p(p) - q_p(p + z) = z^2 A(p, z),$$

where A is positive and C^3 on a neighborhood of the compact core. Uniform strictness of the maximum implies that A is bounded away from zero there. Define

$$\chi_p(z) = z\sqrt{A(p, z)}.$$

Then $q_p(p) - q_p(p + z) = \chi_p(z)^2$. Because A is positive and smooth, $(p, z) \mapsto \chi_p(z)$ is locally one-to-one with derivative bounded away from zero uniformly on the compact core. The reflected endpoint is therefore

$$\ell_p(h) = \chi_p^{-1}(-\chi_p(-h)).$$

The map $(p, h) \mapsto \ell_p(h)$ is C^3 with derivatives bounded uniformly on the compact core. Substitution into the exact gain integral makes the normalized remainder a continuous function of (p, h) after division by h^4 . Compactness then yields a common constant C such that

$$|\Gamma_F(p, h) - K_F(p)h^3| \leq Ch^4$$

for all sufficiently small h , uniformly in p on the core. $\square$

Proposition A.10.1 (Uniform local improvement). *If $K_F(p) \leq -\eta < 0$ throughout C , then for*

$$0 < h \leq h_0 := \min\{\bar{h}, \eta/(2M)\},$$

every admissible reflected excursion based in C satisfies

$$\Gamma_F(p - h, p, p + \ell_p(h)) \leq -\frac{\eta}{2}h^3 < 0.$$

Proof. Combine Theorem A.10.2 and Lemma A.10.4. $\square$

Lemma A.10.5 (Slack-based insertion). *Let a globally compatible ridge-contact family leave a smooth interval containing the support of a faster exact excursion e , and let $\delta = -\Gamma_F(e) > 0$ be the time saved. Suppose every type assigned before the departure state has slack at least $\sigma > 0$ against every post-reentry continuation alternative. If $\bar{U}$ bounds the absolute value of those later dated payoffs and*

$$(e^{r\delta} - 1)\bar{U} < \sigma,$$

then replacing the smooth interval by e preserves global incentive compatibility.

Proof. The excursion returns to the same public state and advances every post-reentry date by δ . Comparisons among post-reentry alternatives are unchanged because every corresponding payoff line is multiplied by the same positive factor $e^{r\delta}$. Comparisons of later assigned types against earlier alternatives are relaxed. The only potentially tighter constraints are those of types assigned before departure against post-reentry alternatives. Each such payoff rises by at most $(e^{r\delta} - 1)\bar{U}$, which is smaller than the maintained slack. $\square$

The lemma explains why a local negative gain is not automatically a global deviation. At an interior smooth boundary, upstream waiting constraints may bind and the slack condition can fail. Global selection therefore uses an explicit replacement condition.

Definition A.10.1 (Replacement condition). Fix a globally compatible excursion e . For any globally compatible family $\mathcal{A}$ omitting e , let $\mathcal{C}_e(\mathcal{A}) \subseteq \mathcal{A}$ be a subfamily whose supports conflict with e . The excursion satisfies a replacement condition if, whenever $\mathcal{A}$ is optimal and omits e ,

(i) $(\mathcal{A} \setminus \mathcal{C}_e(\mathcal{A})) \cup \{e\}$ is globally compatible; and

(ii)

$$\Gamma_F(e) \leq \sum_{a \in \mathcal{C}_e(\mathcal{A})} \Gamma_F(a).$$

Proposition A.10.2 (Conflict-dominance selection). *If an excursion e satisfies a replacement condition, some optimal globally compatible ridge-contact family contains e .*

Proof. Take an optimal family $\mathcal{A}$. If it contains e , there is nothing to prove. Otherwise replace $\mathcal{C}_e(\mathcal{A})$ by e . Compatibility is preserved by the condition, and additivity of gains gives a weakly lower total duration. The replacement is therefore also optimal and contains e . $\square$

A useful primitive sufficient case is supplied by Lemma A.10.5: a candidate wave whose support lies after a region of uniformly slack waiting constraints is globally insertable. Another is a wave at the beginning of the active region, where no upstream waiting constraints exist.

For a compact certified core C , define

$$W(C) = \inf\{\Gamma_F(e) : e \text{ is globally compatible and } C \subseteq I(e)\},$$

and let

$$H(C) = \inf \left\{ \sum_{a \in \mathcal{A}} \Gamma_F(a) : \mathcal{A} \text{ is globally compatible, treats } C, \text{ and no } a \in \mathcal{A} \text{ covers } C \right\}.$$

The phrase “treats C ” means that the family has the same prescribed outside components and replaces the smooth benchmark on the relevant core; this keeps the comparison within a common endpoint and compatibility problem.

Corollary A.10.1 (Whole-core selection). *Suppose the outside components can be held fixed when replacing the treatment of C . If $W(C)$ is attained and $W(C) \leq H(C)$, some optimal globally compatible family contains a single excursion whose support covers C .*

Proof. Take an optimal treatment of the core. If it already contains a covering excursion, the claim holds. Otherwise its gain is at least $H(C)$. Replacing its treatment of C by an attaining covering excursion changes the gain to $W(C) \leq H(C)$ and preserves compatibility by hypothesis. The resulting family is optimal and covers C with one excursion. $\square$

Theorem A.10.3 (Primitive concentration and conditional global batching). *Let C be a compact interior core satisfying the regular reflected-contact hypotheses and $K_F \leq -\eta < 0$. Then every point of C supports sufficiently small negative-gain reflected excursions. If one such excursion has a replacement condition, some optimum contains a wave intersecting C . If, in addition, the whole-core hypotheses of Corollary A.10.1 hold, some optimum contains a wave whose support covers C .*

Proof. The local claim follows from Proposition A.10.1. The global claims follow from Proposition A.10.2 and Corollary A.10.1. $\square$

A.11 The one-sandwich benchmark

A useful deterministic benchmark consists of an initial pool, one smooth separating region, and a terminal pooled tail. This pattern is not an unconditional global theorem: with other density shapes, interior waves may improve on the smooth segment. It remains useful for comparative statics within this benchmark.

Let m_0 and m_T be the entry and exit stocks of the separating region. Along the separating ridge,

$$B^*(m) = \frac{r\nu(m)}{r + \beta m}, \quad \tau'(m) = \frac{1}{\beta m} \left\{ \frac{\beta}{r + \beta m} + \frac{1}{\nu(m)f(\nu(m))} \right\}.$$

The time spent moving beliefs through the separating region is therefore

$$T_{\text{belief}} = \int_{m_0}^{m_T} \frac{1}{\beta m} \left\{ \frac{\beta}{r + \beta m} + \frac{1}{\nu(m)f(\nu(m))} \right\} dm.$$

The remaining delay required to keep adjacent types ordered can be written as an endpoint rent term,

$$T_{\text{rent}} = \frac{1}{r} \log \left[\frac{(\nu(m_0) - B^*(m_0))\nu(m_T)}{(\nu(m_T) - B_T)\nu(m_0)} \right],$$

with the exact normalization depending on the terminal acceptance rule. Two distributions sharing the same active-region boundaries have the same endpoint rent term even if their belief-time integrals differ. This decomposition applies only within the one-sandwich benchmark; additional waves change the endpoints and may improve on it.

A.12 Smooth rationing flows

Let $W(t)$ be continuation value and $h(t)$ service hazard. Dynamic programming over dt gives

$$W(t) = h(t)dt g_v(t) + (1 - h(t)dt)W(t + dt) + o(dt),$$

so

$$W'(t) = h(t)\{W(t) - g_v(t)\}.$$

If the marginal type remains indifferent, $W = g_v$ and $g'_v = 0$. This gives

$$B_t = \frac{rv}{r + \beta M_t}, \quad \dot{M}_t = M_t(r + \beta M_t).$$

If $x(t)$ is remaining mass of the marginal type and no other type is served on the interval, then

$$h(t) = \frac{M_t(r + \beta M_t)}{x(t)}.$$

For a higher type v_i , mimicking payoff is the marginal type's constant payoff plus $(v_i - v)$ times the average discounted service probability. This scalar moment is the flow analogue of the finite s/t lottery in the counterexample.

A.13 Additional free-supply curvature details

For the uniform distribution, write $d = \bar{v} - \underline{v}$ and $v(M) = \bar{v} - dM$. Substituting into the marginal condition gives

$$B(M) = \frac{r(\bar{v} - dM)}{r + \beta M}.$$

The belief law implies

$$\dot{M} = -\frac{\beta M B(M)}{B'(M)}.$$

Direct differentiation shows that $\dot{M} > 0$ on the active range. The sign of $\ddot{M}$ equals the sign of a quadratic expression in M whose derivative is negative under the dispersion condition stated in the main text. Hence the sign changes at most once. Evaluating at the two active boundaries yields the prior threshold ensuring positive initial curvature and negative terminal curvature.

The same calculation also clarifies why within-class convexity does not automatically imply an aggregate S-shape. A discrete finite-type path can contain flat intervals and upward jumps in the adoption rate. The atomless result smooths these transitions by replacing a finite type boundary with a continuum of marginal values.

A.14 The open-middle competitor in the counterexample

Serve all type 1 at zero and release an open middle batch at date s . Because service at s is certain, type 1's no-delay constraint is

$$v_1 - B_0 \geq e^{-rs}(v_1 - B_0 e^{-\beta q_1 s}).$$

The smallest positive binding solution is $s \approx 2.030696703$. At that date $B_s \approx 0.612166030$. Type 2 immediately jumps to the stock satisfying its ridge condition,

$$M^*(B_s) = \frac{r(v_2 - B_s)}{\beta B_s} \approx 0.245468008,$$

and then follows the smooth law to 0.6. The smooth duration is approximately 1.504388580. The additional wait until $B = v_3$ at stock 0.6 is approximately 1.052843165. Total completion is therefore about 4.587928448, well above the four-release construction.

This calculation is useful because it rules out the most natural non-simple three-date objection. An open batch cannot deliver rationing at its release and preserve unused capacity afterward. It must choose between a low-probability early opportunity and certain service; the counterexample benefits from using distinct releases for those functions.

A.15 Terminal padding and asymptotic calendar simplicity

This section separates the number of release dates from the richness of rejection-contingent allocations. The argument does not purify service lotteries. It shows instead that a terminal-only value block can be subdivided into arbitrarily many nearby value classes without changing the operational outcome.

Lemma A.15.1 (Strict terminal-only interval). *Consider an immediately exhausted equilibrium with positive release dates*

$$t_1 < \dots < t_K = T,$$

and acceptance probabilities $Q_k > 0$. Suppose $K \geq 2$ and positive adoption occurs before T . Then $B_{K-1} > B_T$, and every value

$$v \in (B_T, B_{K-1})$$

strictly prefers, at every preterminal private history, to make no preterminal application and to apply at T .

Proof. Beliefs strictly fall between t_{K-1} and T because cumulative adoption is positive, so $B_{K-1} > B_T$. Fix $v \in (B_T, B_{K-1})$. At T , applying gives $Q_K e^{-rT}(v - B_T) > 0$, whereas remaining unserved gives zero.

Consider any preterminal release t_k and any private history at which the agent remains eligible. Let $W_k(v) > 0$ be the continuation value from declining the current opportunity and following the terminal-only strategy thereafter. If the agent applies at t_k , receives service with probability Q_k , and after rejection returns to that continuation, utility is

$$Q_k e^{-rt_k}(v - B_k) + (1 - Q_k)W_k(v).$$

Because $B_k \geq B_{K-1} > v$, the immediate-service term is negative. Hence

$$Q_k e^{-rt_k}(v - B_k) + (1 - Q_k)W_k(v) < W_k(v).$$

Deleting the current application is therefore strictly profitable. Backward induction removes every preterminal application. Applying at T is strictly optimal because $v > B_T$. The argument applies after every rejection history because rejection preserves eligibility and the public continuation is unchanged by an atomless individual's action. $\square$

Proposition A.15.1 (Exact terminal padding). *Suppose a finite immediately exhausted equilibrium has N distinct values, K positive releases, and a positive mass z of terminal-only applicants whose values lie in an open interval $I \subset (B_T, B_{K-1})$. For every integer $L \geq 1$ and every $\eta > 0$, there is a finite economy within Prokhorov distance η of the original distribution, with at least $N + L$ distinct values, in which the same supply plan has an equilibrium generating exactly the same applicant masses, acceptance probabilities, service path, beliefs, and completion time.*

Proof. Choose a positive submass $0 < \delta < \min\{z, \eta\}$ of the terminal-only block and partition it into positive masses $\delta_1, \dots, \delta_L$. Select L distinct values $w_1, \dots, w_L \in I$ within η of values in the original terminal-only block, and replace the chosen submass by classes (w_ℓ, δ_ℓ) . Prescribe terminal-only application to every new class.

By Lemma A.15.1, terminal-only application is strictly optimal for each w_ℓ at every private history. The total mass following every operational strategy is unchanged. Thus applicant masses and acceptance probabilities at every release are unchanged, as are service, adoption, exposure, beliefs, and completion. Every original type faces exactly the old environment and remains optimizing. The distributional perturbation is bounded by the relabeled mass and value displacement and can be made smaller than η . $\square$

Taking $L \geq \max\{0, K - N\}$ makes the unchanged K -release plan simple in the padded economy. The proposition is exact for outcomes, but it does not claim that the unchanged plan is optimal after padding.

Definition A.15.1 (Terminal-rich foundation). A sequence of finite operational equilibria is terminal-rich if, for all sufficiently large indices, it contains a positive terminal-only block in an open interval $I_N \subset (B_{T_N}, B_{K_N-1})$, and the mass relabeled in the padding construction can be chosen to converge to zero.

Theorem A.15.1 (Asymptotic value-density of simple calendars). *Let $F_N \Rightarrow F$ be a terminal-rich sequence of finite economies with immediately exhausted equilibrium outcomes π_N , completion times $T_N \rightarrow T$, and outcomes converging to a relaxed operational outcome for F . Let N_N and K_N be the numbers of values and releases. Then there exist padded economies $\tilde{F}_N \Rightarrow F$ and simple equilibria $\tilde{\pi}_N$ that generate exactly the same aggregate outcomes and satisfy $T(\tilde{\pi}_N) = T_N$.*

If, in addition, T is the relaxed operational value $V^R(F)$, and if bounded-completion finite outcomes are sequentially compact in the relaxed operational topology described in Section A.16, then the unrestricted and simple-plan values $\tilde{V}_N$ and $\tilde{V}_N^S$ in the padded economies satisfy

$$\tilde{V}_N \rightarrow V^R(F), \quad \tilde{V}_N^S \rightarrow V^R(F), \quad \tilde{V}_N^S - \tilde{V}_N \rightarrow 0.$$

Proof. Apply Proposition A.15.1 with

$$L_N = \max\{0, K_N - N_N\}$$

and with relabeled mass and value displacement converging to zero. The padded distributions converge to F , while the equilibrium outcomes and completion times are unchanged. Since the number of values is at least the number of positive exhausted releases, each padded plan is simple.

Let $\tilde{V}_N$ and $\tilde{V}_N^S$ denote the unrestricted and simple-plan completion values in the padded economies. Feasibility of the padded equilibrium gives

$$\tilde{V}_N \leq \tilde{V}_N^S \leq T_N.$$

Suppose along a subsequence that $\tilde{V}_N \leq V^R(F) - \varepsilon$ for some $\varepsilon > 0$. The displayed upper bound provides a common completion bound. Relaxed operational compactness then yields a subsequential limit feasible for F with completion no greater than $V^R(F) - \varepsilon$, contradicting the definition of $V^R(F)$. Hence

$$\liminf_N \tilde{V}_N \geq V^R(F).$$

Together with $T_N \rightarrow V^R(F)$ and the sandwich above, this proves convergence of both values and of their difference. $\square$

Remark 4 (Scope). The theorem is an asymptotic calendar result. It does not show that the padded equilibrium is an exact finite optimum, that every relaxed operational optimum is terminal-rich, or that lottery-valued limits reduce to deterministic schedules. Those are separate questions. For the deterministic batch-smooth-batch schedule, the positive terminal batch supplies the required terminal-only interval away from its cutoff, so any operational foundation reproducing that block is terminal-rich.

A.16 Closure under service-time lotteries

Let Λ_N be the joint measure on value and service time induced by the finite outcome, with first marginal bounded by the value distribution. Tightness follows from the compact value support and the common completion bound. Prokhorov's theorem gives a weakly convergent

subsequence. The induced cumulative adoption path is the time marginal of Λ_N and converges in completed-graph topology.

For any bounded continuous test function $\phi(v, t)$,

$$\int \phi d\Lambda_N \rightarrow \int \phi d\Lambda.$$

The payoff kernel $e^{-rt}(v - B_t)$ is continuous once uniform belief convergence is established. Incentive constraints compare integrals of this kernel against report-dependent service measures; weak convergence therefore preserves every finite pairwise inequality. A monotone-class argument extends closure from a countable dense set of values to almost every type, and upper semicontinuous envelopes restore the remaining null set without changing aggregate service.

Lower semicontinuity of completion follows because if every approximating path reaches $\bar{M}$ by T_N and $T_N \rightarrow T$, then the completed graph of the limit contains a point with adoption at least $\bar{M}$ no later than T . This argument permits terminal jumps.

Disintegrating the limit joint measure gives service-time kernels $K_v(dt)$. A schedule-valued limit is the special case $K_v = \delta_{\tau(v)}$ for almost every served value. Weak compactness alone does not imply this degeneracy: a positive set of values may retain nondegenerate rejection-contingent service lotteries even when every finite value atom vanishes. Terminal padding likewise changes the number of value labels without changing the service lotteries in the underlying outcome.

Accordingly, define $V^R(F)$ as the infimum completion time over relaxed operational limits and $V^D(F)$ as the infimum over ordered deterministic direct schedules. These are different optimization domains unless an implementation theorem connects them. The deterministic results characterize $V^D(F)$ sharply under the stated density conditions; they do not establish $V^R(F) = V^D(F)$. If the canonical deterministic schedule is itself an operational limit, then $V^R(F)$ cannot exceed its completion time. The finite three-type construction demonstrates that rationing can have screening value with atoms, but it neither proves nor disproves a strict relaxed advantage for a smooth atomless distribution.